

Chemotherapeutic Agents

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Drug Classification

Drugs can be classified according to their medicinal uses into two main classes:

I-Pharmacodynamic agents: Drugs that act on the various physiological functions of the body (e.g. general anaesthetic, hypnotic and sedatives, analgesic etc.).

II-Chemotherapeutic agents: Those drugs which are used to fight pathogenic (e.g. sulphonamides, antibiotics, antimalarial agents, antiviral, anticancer etc.).

Antiseptics and disinfectants

- Synthetic antibacterial compounds are divided into two major classes: **topical agents (germicides)** and **systemic agents**.
- The topical agents are termed disinfectants, antiseptics, and preservatives, depending on how they are used.
- **Antiseptics** are compounds that kill or prevent the growth of micro organisms when applied to living tissue.
- **A disinfectant** is an agent that prevents transmission of infection by the destruction of pathogenic microorganism when applied to inanimate objects.
- Antiseptic in higher concentration can be used as disinfectants.
- Antiseptics and disinfectants differ from systemic agents in that they show **little selective toxicity** between the microbes and the host.

General uses

Disinfectants

- In hospital sanitation, including sterilization of surgical instruments, public health methods, and in the home.

Antiseptics

- Have important applications in the preoperative preparation of both surgeons and patients.
- They also are used in treating local infections caused by microorganisms refractive to systemic antimicrobial agents.

Other commonly used terms

- **Sanitizer** - refers to an agent that reduces the number of harmful bacteria to an established safe limit.
- The criterion for sanitization is killing of 99.999% of specific test bacteria in 30 s, and it is commonly applied to eating and drinking implements and dairy equipment.
- **Asepsis**-means the prevention of contamination by microorganisms;
- **Decontamination**- refers to the disinfection or sterilization of infected things;
- **Sterilization**- refers to killing all forms of microorganisms;
- **Fumigation** is the exposure of an area or object to disinfecting fumes.

Ideal characteristics of a disinfectant

- Rapid lethal action against all pathogens
- Good penetration in to organic matter
- Compatible with organic compounds (especially soaps)
- Non-corrosive
- Non-staining, odorless and aesthetically acceptable

Ideal characteristics of antiseptics

- Rapid and sustained lethal action against all pathogens
- Low surface tension
- Remain active in presence of body fluid
- Non-irritating and non-allergenic
- Lack systemic toxicity

- **Classes of antiseptics and disinfectants**
 - alcohols and related compounds
 - aldehydes
 - phenols and their derivatives
 - oxidizing agents
 - halogen containing compounds
 - cationic and anionic surfactants
 - dyes
 - mercury compounds

- **Alcohols and related compounds**

- Alcohols have desirable properties as disinfectants and antiseptics.
- They are bactericidal, inexpensive, and relatively nontoxic when used topically.
- They are more active against vegetative forms than against spores.
- Two of most commonly used antiseptic and disinfectants are ethyl and isopropyl alcohol.

SAR

1. The antibacterial potencies of primary alcohol (against *S.aures*) increase with molecular weight until the 8-carbon atom octanol reached.
 - As the primary alcohol chain length increases, van der Waals' interaction increase and the ability to penetrate microbial membranes increases
 - As water solubility decrease the apparent antimicrobial potency diminishes with molecular weight.

2. Branching of the alcohol chain decrease antibacterial potency

- Weak van der Waals' forces brought about by branching do not penetrate bacterial cell membrane as efficiently.
- The isomeric alcohols' potencies decrease in order of
- $1^\circ > 2^\circ > 3^\circ$ alcohol
- Despite this fact, 2 propanol is (isopropyl alcohol is used commercially instead of n-propyl alcohol, b/c it is less expensive.

MOA

Denature important proteins and carbohydrates

Commercially available alcohols

Alcohol, USP. Ethanol

- Used as antiseptic, preservative, mild counter irritant and as a solvent.
- Germicidal action occurs with conc. of 60-90%
- Metabolized in human body by series of oxidation

Isopropyl alcohol (isopropanol)

- prepared commercially by the **sulphuric acid catalyzed hydration of propylene**
- bactericidal in the concentration range of 50-95%

Ethylene oxide

- diffuse readily through porous materials and very effectively destroys all forms of microorganisms at ambient temperature

Halogen containing compounds

- **Chlorine and Chlorophores**
 - Chlorine has been used since 1827 to disinfect drinking water and it remains the leading agent for this purpose because it is **cheap** and **effective**.
 - Chlorophores and related compounds have been used in drinking water. They includes
- **A) Chlorine dioxide (ClO₂)**
- It is prepared at the site of consumption by treating a **sodium chlorite** solution with **chlorine**
- Used in recent years for **drinking water disinfection** and **wastewater treatment**



B) Inorganic chloramines (ClNH_2 , Cl_2NH , and Cl_3N)

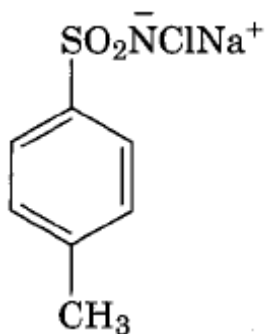
- Used to improve the taste of drinking water
- inferior disinfectant properties
- More recently, the addition of inorganic chloramines to water is being reconsidered because they prevent the formation of **carcinogenic trihalomethanes** from pollutants

C) Inorganic hypochlorites

- They include **sodium hypochlorite**, **calcium hypochlorite dihydrate**, **lithium hypochlorite**, and **chlorinated trisodium phosphate** [$\text{NaOCl} \cdot 4(\text{Na}_3\text{PO}_4)$]
- They release **hypochlorous acid** in their water solution which is responsible for their germicidal activity.

D) Organic N-chloro derivatives

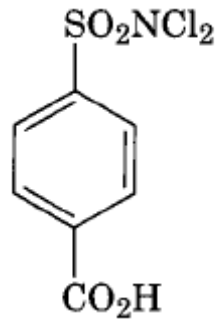
- Produced in bulk quantities by treating appropriate nitrogen compounds with HOCl
- In water they slowly release HOCl and regenerate the parent nitrogen compound
- Some examples of such compounds
- **Chloramine-T (sodium p-toluenesulfonchloramide)**



used for disinfecting wounds

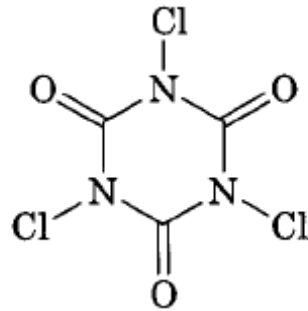
Halazone (p-dichlorosulfamoylbenzoic acid)

➤ Used to disinfect drinking water



Trichloroiso cyanuric acid (trichloro-s-triazinetriene)

- Widely used to disinfect swimming pools

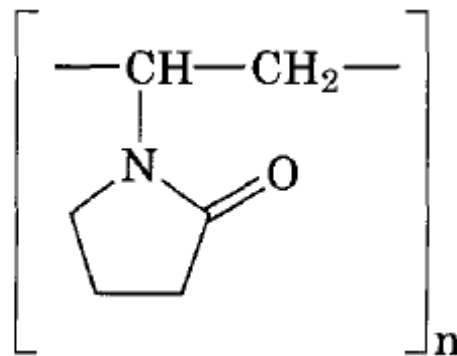


- **Chlorine** in minute amounts results in a destructive **permeability of bacterial cell walls**
- Two different mechanisms have been proposed for the germicidal effects of **hypochlorous acid**.
 - oxidation of sulfhydryl groups in the proteins
 - chlorination of amide nitrogens

2. Iodine

- Molecular iodine (I_2) is one of the oldest germicides and still one of the most useful.
- It is highly effective, economical, and minimally toxic at the usual concentrations.
- Water solutions of iodine are complex systems containing at least seven species, of which I_2 and $HI0$ are considered to be the main disinfectants.
- Iodine preparations include the **tincture** (2% solution of I_2 in 50% ethanol containing iodide), **Lugol's** solution (5% I_2 in water with KI), and iodine solution (2% I_2 in water with NaI).

- The suggested **mechanisms of action** include **oxidation of the SH groups in proteins, iodination of amino groups, iodination of tyrosine residues, and addition to the double bonds of unsaturated fatty acids.**
- Povidone-iodine is a complex of I_2 with polyvinylpyrrolidone, a nonionic surfactant polymer.
- This water-soluble complex contains approximately 10% of available I_2 which is released slowly.
- Povidone- iodine is nontoxic, nonirritating, and nonstaining.
- It is used as an antiseptic before surgery and injections and for treating wounds and lacerations.



Cationic and anionic surfactants

- **Cationic surfactants**
- Cationic surfactants are quaternary ammonium or pyridinium salts that are ionic in water and have surface-active properties.
- Cationic surfactants show potent activity against Gram-positive bacteria and lower activity against Gram-negative bacteria.
- The **mechanism of action of cationic surfactants** is association with **cell wall protein** followed by **penetration and disruption of the cell membrane**
- **Desirable features of cationic surfactants** include water solubility, low toxicity, relatively good tissue penetration, and freedom from stains and corrosion.

Disadvantages include

- inactivation by anionic surfactants (all traces of soap must be removed from skin),
- reduced effectiveness in the presence of blood serum
- strong adsorption on fibrous material such as cotton,
- occasional allergic responses on prolonged use, and resistant organisms.

The important surfactants are

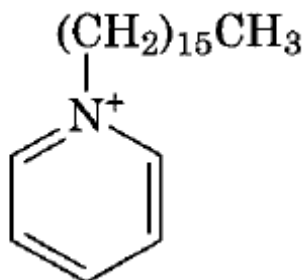
- benzalkonium chloride,
- cetylpyridinium chloride.

Benzalkonium chloride

- Benzalkonium chloride is a mixture of alkylbenzyldimethylammonium chlorides having the general formula $[\text{C}_6\text{H}_5\text{CH}_2\text{N}(\text{CH}_3)_2\text{R}]^+ \text{Cl}^-$, where R is a mixture of alkyl groups of which $\text{C}_{12}\text{H}_{25}$, $\text{C}_{14}\text{H}_{29}$, and $\text{C}_{16}\text{H}_{33}$ are the main components.
- It is used as an antiseptic for skin and mucosa in concentrations of 1:75 to 1:20,000.
- Other uses include irrigation and disinfection of surgical instruments.

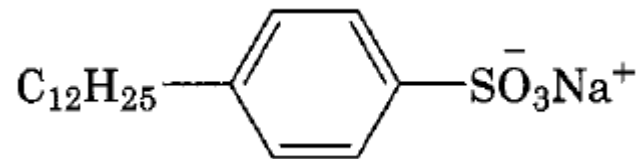
Cetylpyridinium chloride (1-hexadecylpyridinium chloride)

- Has positively charged nitrogen as part of a pyridine ring.
- It finds use as a **general anesthetic**, irrigant for mucous membranes, and **as** a component of **mouthwashes** and **lozenges**.
- Antibacterial potency depends on the electronegativity of substituents on the pyridine ring with electron-releasing groups affording the highest potencies.



Anionic Surfactants

- Mixtures of anionic surfactants with acids to lower the pH to **2-3** show rapid germicidal activity
- **Alkyarylsulfonates** such as **dodecylbenzenesulfonic acid** are the most effective surfactants and **phosphoric acid** is frequently used in the mixture.



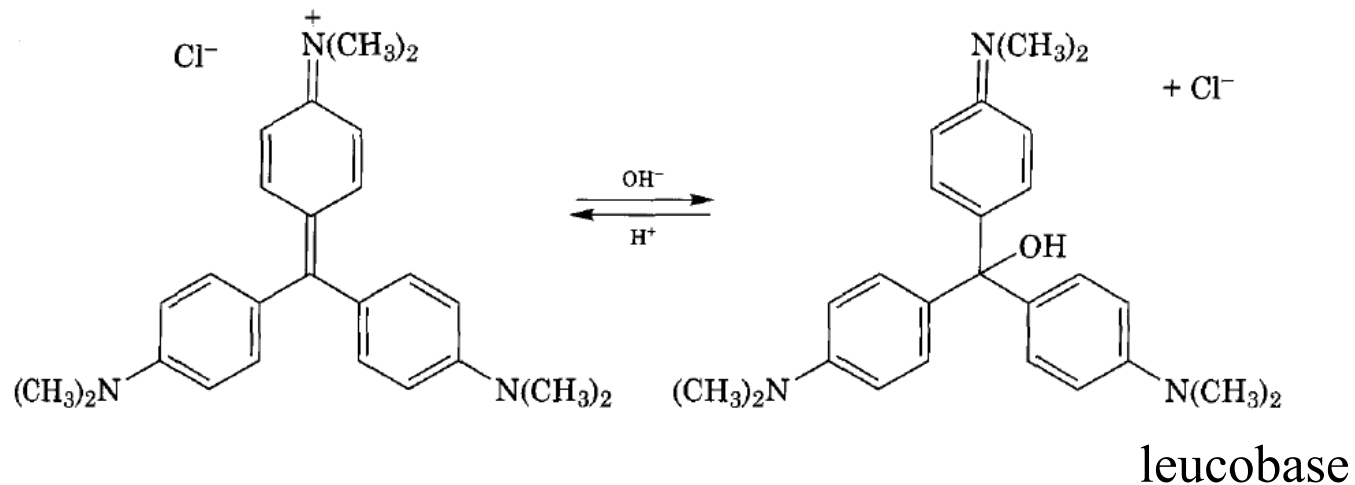
- Possible modes of action are **disorganization of the cellular membrane, denaturation of key enzymes and other proteins, and interruption of cellular transport**
- Products have been developed for disinfecting equipment in the food and dairy processing industry

Dyes

- Organic dyes were used extensively as antimicrobial agents before the development of **sulfonamides** and other **antibiotics**.
- The commonly used dyes now a day are gentian violet and methylene blue.

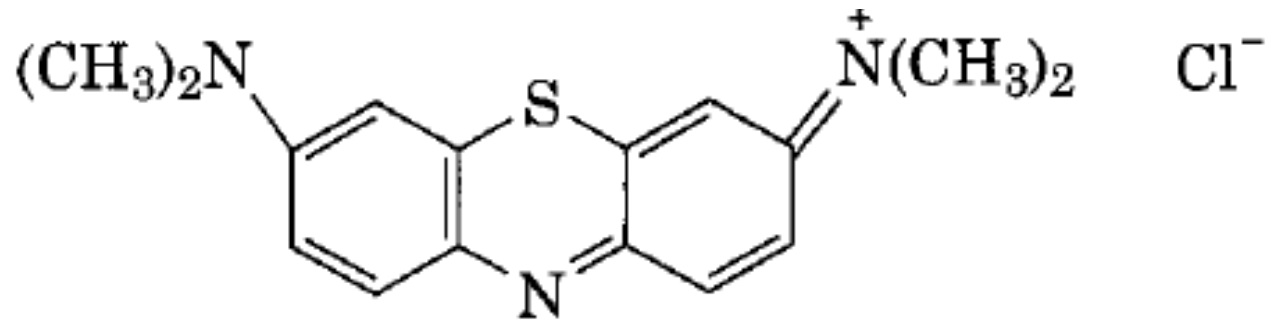
Gentian violet (hexamethyl-p-rosaniline chloride)

- It is a triphenylmethane dye that is converted into a colorless form (leucobase) in alkaline solution
- It has been used as a topical antibacterial agent, but its main use is as a topical agent for fungal infections.



Methylene blue

- It is a bacteriostatic agent that has been used for cystitis and urethritis, infections associated with *E. coli* and *Neisseria gonorrhea*, respectively.

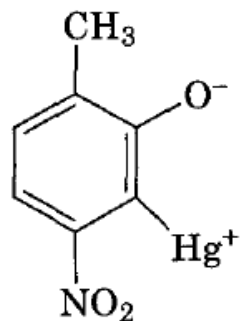


Heavy Metals

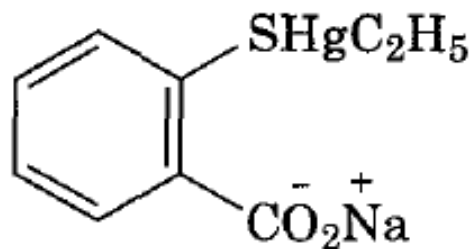
Mercury Compounds

- Ammoniated mercury ($\text{Hg}(\text{NH}_2)\text{Cl}$), which is used for skin infections.
- It is formulated as ammoniated mercury ointment.
- There **are** two basic types of organomercurials: those in which the mercury is covalently bonded to carbon and those in which the mercury is bonded to a heteroatom such as oxygen, *sulfur*, or nitrogen.
- Organomercurials **are** more bacteriostatic, less toxic, and less irritating than inorganic mercury compounds.
- Their mechanism of **action** is thought to result from binding with thiols in enzymes and other proteins.

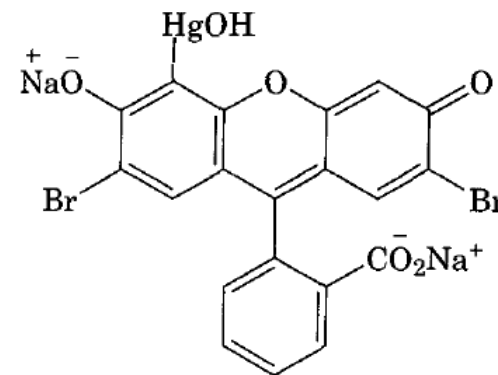
Examples of organomercurials



Nitromerso



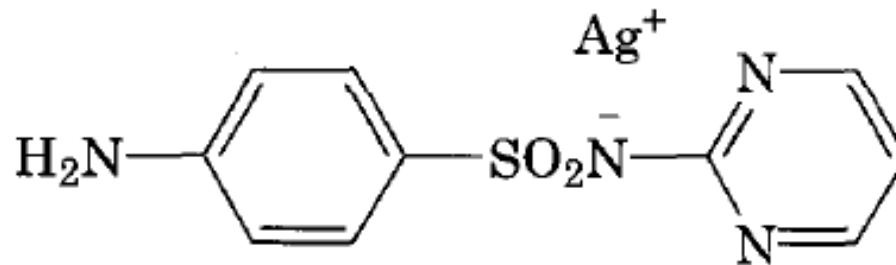
Thimerosal (merthiolate)



Merbromin
(mercurochrome)

Silver Salts

- Silver ions bind readily with biologically important functional groups including **thiol, amine, phosphate, and carboxy- precipitate proteins or chloride ions.**
- Silver nitrate (AgNO_3) solutions are highly germicidal, destroying most microorganisms at 0.1% concentration.
- **Silver sulfadiazine** is used in the topical treatment of extensive burns.
- It is effective against *Pseudomonas aeruginosa*.



Aldehydes

1. Formaldehyde solution (formalin)

- Contains not less than 37% w/v of formaldehyde (HCHO) with methanol
- Readily undergoes oxidation and polymerization, lead to formic acid and Para formaldehyde

MOA

- Believed to involve direct non-specific alkylation of nucleophilic functional groups

Phenols and their derivatives

- Phenol serves as a standard to compare the activity of most germicidal compounds
- **Phenol coefficient** is defined as the ratio of a dilution of a given test disinfectant to the dilution of phenol that is required to kill a strain of salmonella typhi under controlled time and temperature conditions.
- A number of phenols are more bactericidal than phenol itself

SAR

- Substitution with alkyl, aryl and halogen, primarily in p-position, increase bactericidal activity.
- Straight chain alkyl groups enhance bactericidal activity more than branched groups
- Alkylated phenols and resorcinol are less toxic than the parent compound while retaining bactericidal concentration
- Biphenols generally has greater potency than monophenols
- e.g. Hexachlorophene
- Increasing degree of chlorination also increases the potency

Oxidizing agents

- Their germicidal activity is dependent up on their ability to liberate oxygen in the tissues
- Many of them are inorganic compounds (e.g. H_2O_2 , metal peroxides, sodium perborate)
- All of them generate oxygen and oxygen radicals
- Usually they are effective against anaerobic bacteria and can be used in cleansing contaminated wounds

Limitations

- Poor penetration power

Examples

- Carbamide peroxide topical solution
- A complex of urea and hydrogenperoxide ($\text{H}_2\text{NCONH}_2 \cdot \text{H}_2\text{O}_2$)
- When mixed with water hydrogen peroxide is librated
- Used for treatment of oral ulceration in the dental care

Preservatives

- Preservatives are added to various dosage forms and cosmetic preparations to prevent microbial contamination.
- In parenteral and ophthalmic preparations, used to maintain sterility in the event of accidental contamination during use.

Ideal characteristics of Preservatives

- Effective at low concentrations against all possible microorganisms,
- Nontoxic and compatible with other constituents of the preparation,
- stable for the shelf life of the preparation

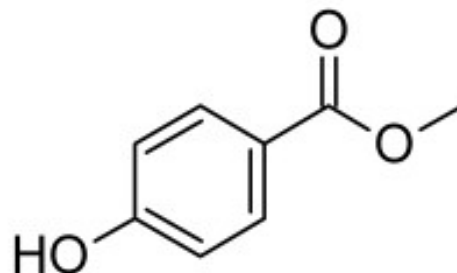
p-Hydroxybenzoic Acid Derivatives

- Esters of *p*-hydroxybenzoic acid (parabens) have distinct antifungal properties.
- Having rapid hydrolysis in vivo to *p*-hydroxybenzoic acid (low toxicity).
- Preservatives for liquid dosage forms
- The preservative activity generally increases with molecular weight, but the methyl ester is most effective against molds, whereas the propyl ester is most effective against yeasts.

- Lipid-soluble propyl ester is the preferred preservative for drugs in oil or lipophilic bases

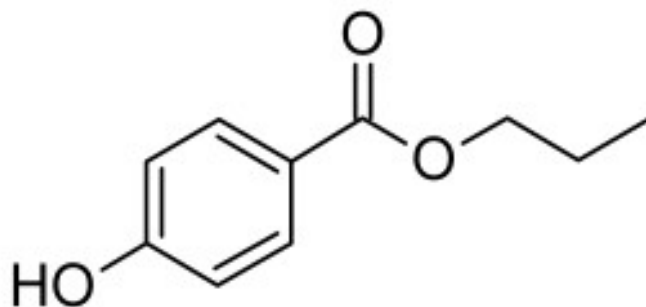
Methylparaben

- Methyl p-hydroxybenzoate(white crystalline powder)
- It is soluble in water and alcohol only slightly soluble in nonpolar organic solvents
- used as a safeguard against mold growth



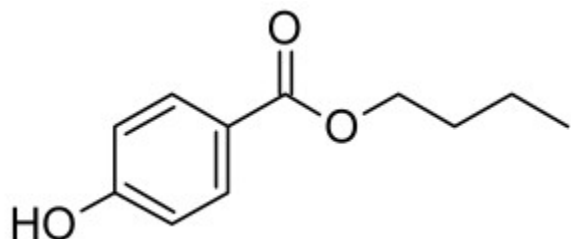
Propylparaben (*p*-hydroxybenzoate)

- white crystalline powder
- slightly soluble in water but soluble in most organic solvents
- Used to retard yeast growth



Butylparaben (*n*-Butyl *p*-hydroxybenzoate)

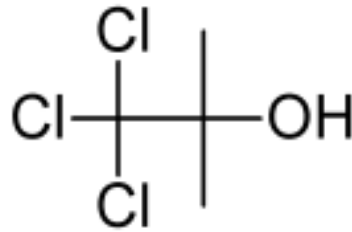
- sparingly soluble in water but very soluble in alcohols and in nonpolar organic solvents



Other Preservatives

Chlorobutanol (1,1,1-Trichloro-2-methyl-2-propanol)

- Used as a bacteriostatic agent in pharmaceuticals for injection, ophthalmic use, and intranasal administration.



Benzyl Alcohol

- Benzyl alcohol (phenylcarbinol, phenylmethanol) occurs naturally

- Commonly used as a preservative **in vials of injectable drugs** in concentrations of 1% to 4% in water or saline solution.
- has the added advantage of having a **local anesthetic action**
- Commonly used in **ointments** and **lotions** as an antiseptic in the treatment of various pruritic skin conditions

Phenylethyl Alcohol

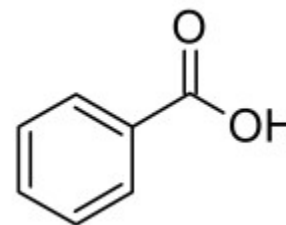
Phenylethyl alcohol (2-phenylethanol, orange oil, rose oil, $C_6H_5CH_2CH_2OH$)

Benzoic acid

Benzoic acid and its esters occur naturally in gum benzoin and in Peru and tolu balsams.

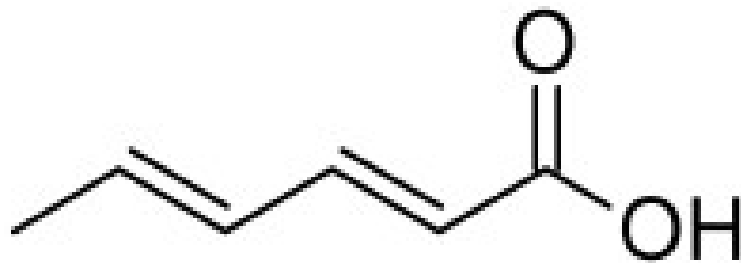
used externally as an antiseptic in lotions, ointments, and mouthwashes

It is more effective as a preservative in foods and pharmaceutical products at low pH



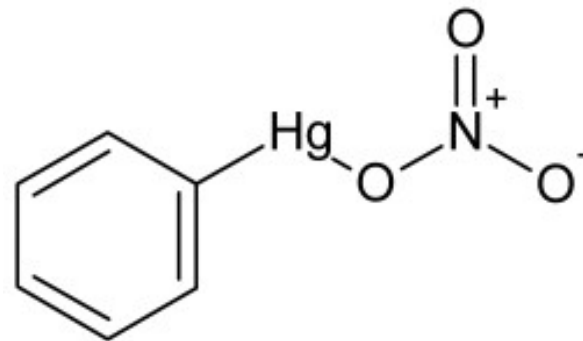
Sorbic Acid

- 2,4-Hexadienoic acid is an effective antifungal preservative.
- used to preserve syrups, elixirs, ointments, and lotions containing components such as sugars that support mold growth.



Phenylmercuric Nitrate

- A mixture of phenylmercuric nitrate and phenylmercuric hydroxide.
- It is used in concentrations of 1:10,000 to 1:50,000 to preserve injectable drugs against bacterial contamination.
- bacteriostatic efficacy is reduced in the presence of serum.



Antibacterial Agents

History

- 1929- Penicillin was discovered by Alexander Fleming
- 1935- Discovery of sulfonamides, synthetic antimicrobials
- Penicillins were isolated and characterized in the mid-1940s
- During this time penicillin G was proved to be effective in curing infection in animals with low toxicity
- 1940s and 1950s a number of other antibiotics including tetracycline, chloramphenicol, and streptomycin were discovered
- 1962- Nalidixic acid, the lead compound for quinolones, was discovered

Classification of Antibacterial Agents

Based on Mechanism of Action

Antimetabolites

- Sulfonamides, Trimethoprim, Pyrimethamine

Cell-wall synthesis inhibitors

- b-lactams (penicillins, cephalosporins, carbapenems, monobactams)
- Vancomycin

Protein synthesis inhibitors

- Tetracyclines (e.g. doxycycline, minocycline, tetracycline, tigacycline...etc)
- Aminoglycosides (e.g. Kanamycin, Gentamicin, Tobramycin)
- Macrolides (erythromycin, clarithromycin, Azithromycin)

Classification

Cont.....

- Ketolides (e.g. telithromycin)
- Lincosamides (e.g. clindamycin)
- Streptogramins
- Oxazolidinones (e.g. Linezolid)
- Miscellaneous (e.g. chloramphenicol)

Nucleic acid synthesis inhibitors

- Quinolones (e.g. ciprofloxacin, levofloxacin, gatifloxacin)

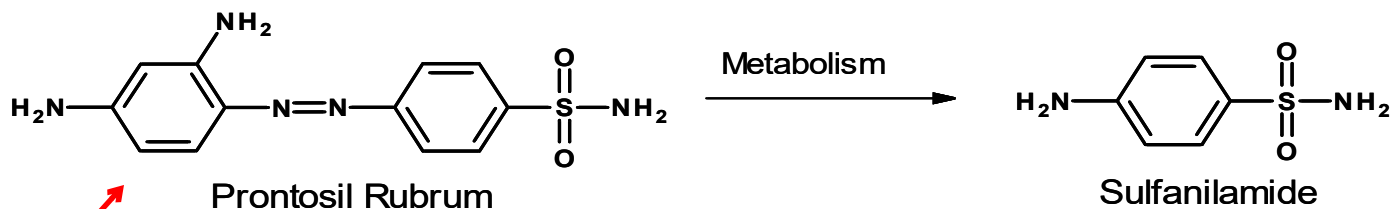
1. Antimetabolites

- The drug inhibits a normal metabolic process by acting as false substitutions
- The drugs masquerade as the natural precursor compound and are often incorporated or compete with the endogenous molecule for the same protein target
- The antimetabolite antibacterial agents include sulfonamides and trimethoprim

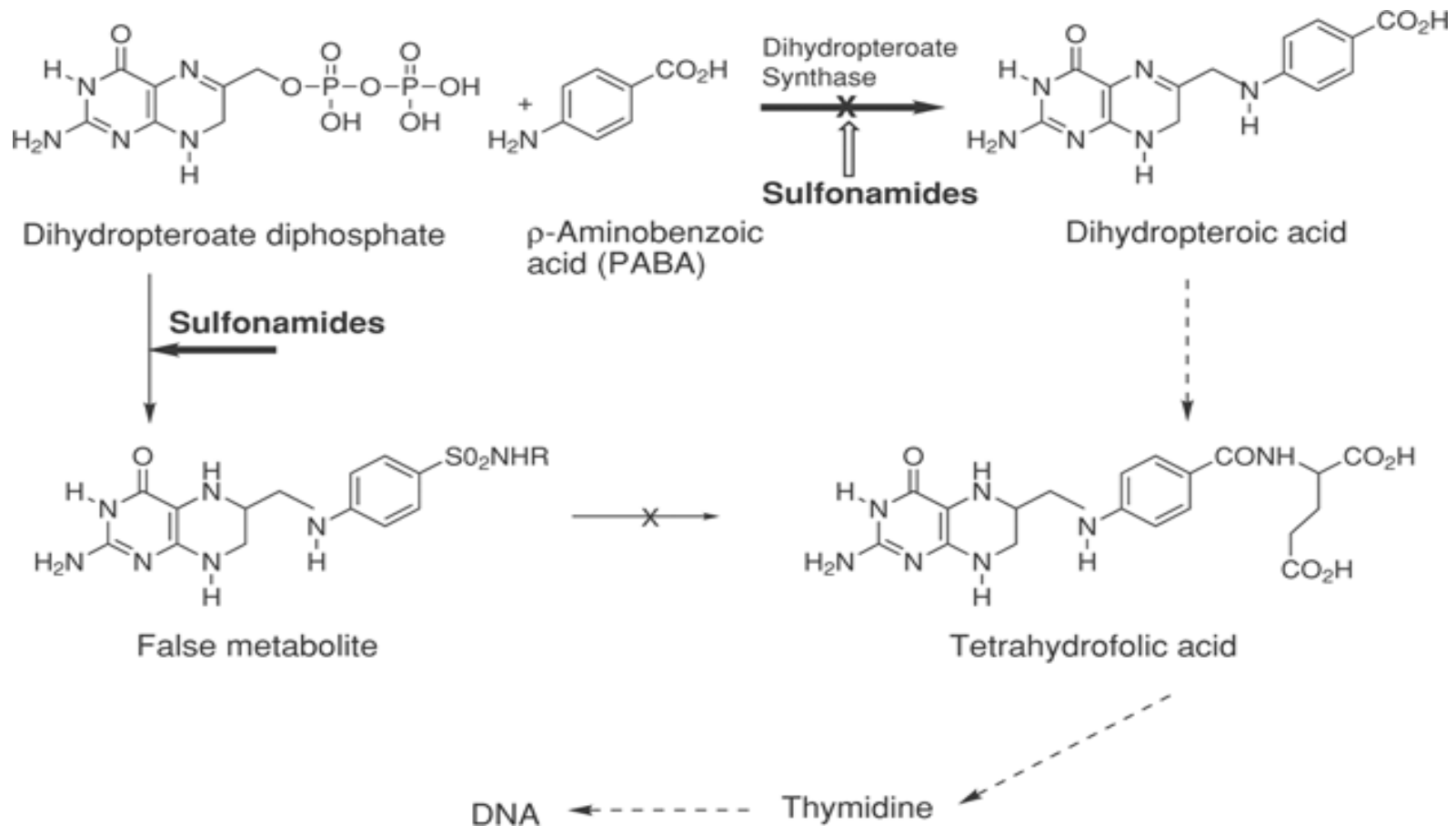
Sulfonamides

History

- In 1932, Prontosil rubrum, was one of a number of dyes examined by Gerhard Domagk for possible selective uptake by certain pathogenic bacteria but not by human cells
- it was not active *in vitro* (i.e. in bacterial cultures)
- Was found to be active *in vivo* against streptococcal infections in mice
- It was demonstrated that the urine of prontosil rubrum–treated animals was bioactive *in vitro*
- Isolation and characterization of the components in the urine led to identification of the active substance as p-aminobenzenesulfonic acid amide (sulfanilamide)



Mechanism of Action of Sulfonamides

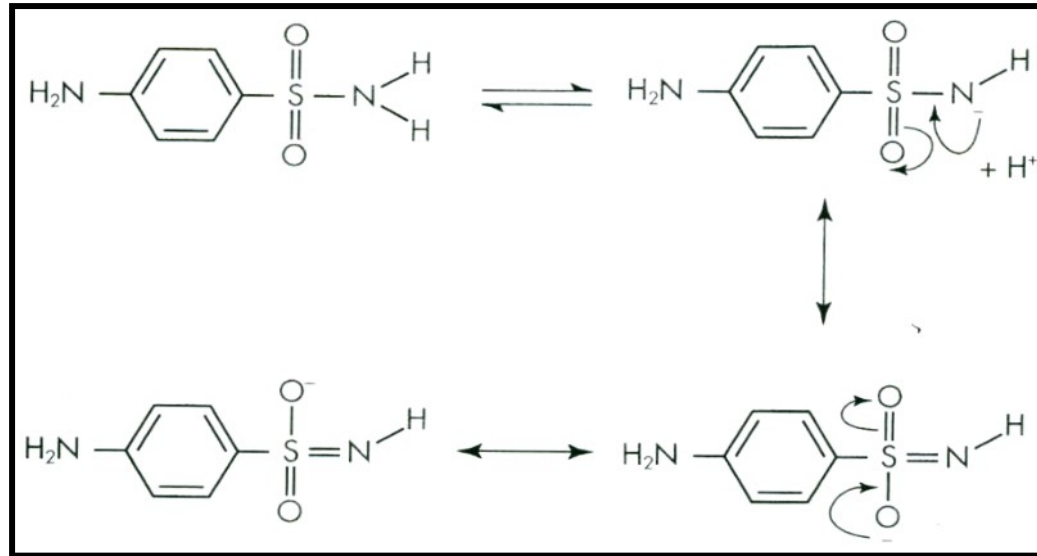


Microbial biosynthetic pathway leading to tetrahydrofolic acid

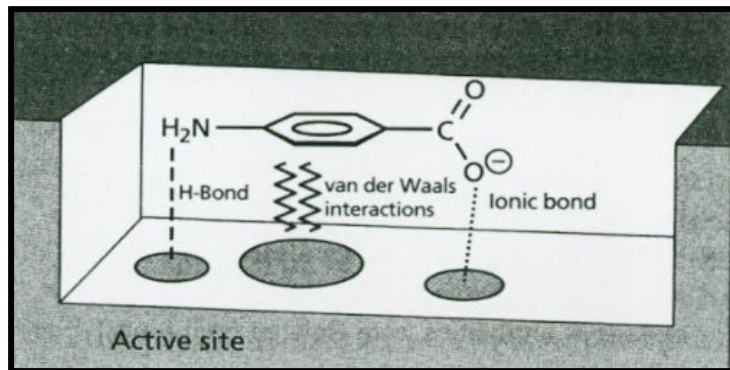
Mechanism of Action (continued)

- Sulfonamides are bacteriostatic
- inhibit the enzyme dihydropteroate synthase
- dihydropteroate synthase is an important enzyme that catalyzes the first step in the synthesis of tetrahydrofolic acid (THF)
- THF is essential in the synthesis of thymidine and other biosynthetic reactions that require transfer of one carbon
- Sulfonamides compete at the active site with p-aminobenzoic acid (PABA), a precursor in the synthesis of folic acid derivatives
- sulfonamides are also incorporated to the dihydropteroate diphosphate in the place of the normal PABA
- The resulting unnatural product is not capable of undergoing the next necessary reaction
- The net result is inability of the bacteria to multiply

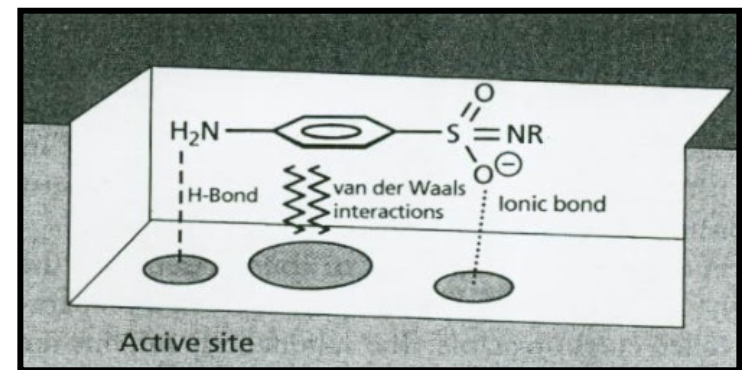
Binding of PABA and Sulfonamide to Dihydropteroate Synthase



Ionization of sulfonamides



Dihydropteroate synthase/
PABA



Dihydropteroate synthase/
Sulfonamide

Mechanism of Action and Selectivity

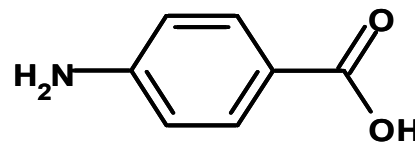
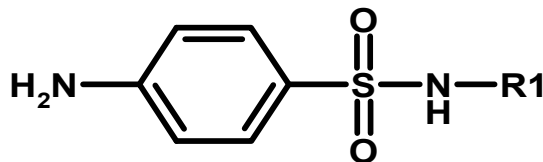
- Humans do not synthesize folic acid because we lack the necessary enzymes, including dihydropteroate synthase
- Instead we get folic acid from our diet
- As a result, sulfonamides have no similar lethal effect on human cell growth
- Therefore, sulfonamides are selective against microbial cells

Bacterial Resistance to Sulfonamides

- Some bacterial species acquire resistance by synthesizing more PABA
- Mutations in dihydropteroate synthase that decrease its affinity for sulfonamides.
- Decreased permeability of the cell membrane to the sulfonamide
- Utilizing a "salvage pathway" which bypasses 1-carbon synthesis of nucleic bases

Structure-Activity Relationship (SAR) of Sulfonamides

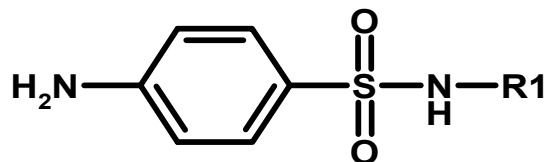
- The sulfonamide moiety of sulfanilamide is weakly acidic due to the strongly electron-withdrawing character of the aromatic SO_2 ($\text{pK}_a = 10.4$)
- The pK_a of the carboxyl group of PABA is about 6.5
- It was shown that substitution of one of the NH_2 hydrogens with an electron-withdrawing heteroaromatic ring greatly increased acidity and dramatically enhanced antimicrobial potency
- With an appropriate substituent the pK_a is reduced to the same range as that of PABA
- Enhancement of the acidity of the sulfonamide increases the antimicrobial potency as well as water solubility under physiological conditions



PABA

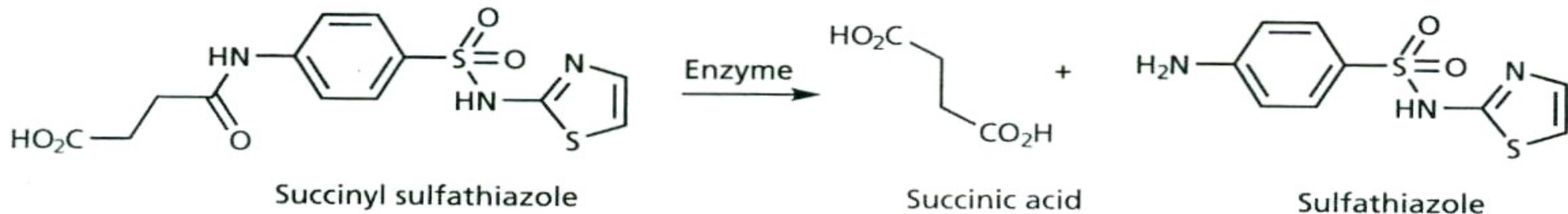
SAR of Sulfonamides

- The anilide amine and sulfonamide functional groups must be *para* to each other
- The *para*-amine (anilide amine) must be unsubstituted (i.e. R-NH₂), unless the molecule is designed to be a prodrug
- The aromatic ring and sulfonamide functional groups are both required for activity and must be attached to the aromatic ring
- The aromatic ring must be *para*-substituted only; extra substitution eliminates activity for steric reasons
- The sulfonamide nitrogen must be primary (R-NH₂) or secondary (R₂-NH)
- R1 is the only possible site that can be varied (except when designed as prodrugs)



SAR of Sulfonamides

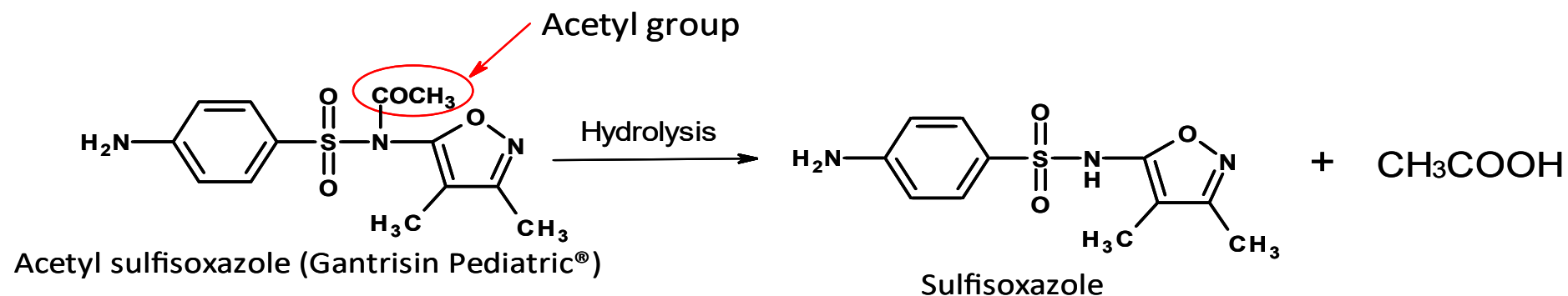
- Anilide amine must be unsubstituted (i.e. R-NH₂) to ensure an active sulfonamide
- However, amides can be synthesized as prodrugs
- Although the amide derivative are inactive, they can be metabolized to regenerate an active unsubstituted compound
- Succinyl sulfathiazole is prodrug of sulfathiazole



Therapeutic Application

Acetyl sulfisoxazole (Gantrisin Pediatric®)

- Is a prodrug of sulfisoxazole
- The acetyl derivative of sulfisoxazole is tasteless and, therefore, suitable for oral administration, especially in liquid preparations
- The prodrug is hydrolyzed in the intestinal tract to give the active drug sulfisoxazole
- Used for the treatment of urinary tract infection (UTI) and otitis media



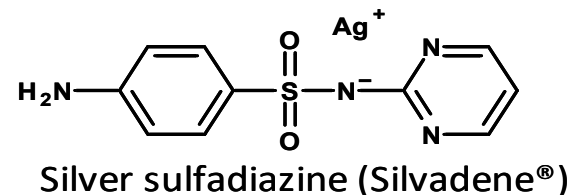
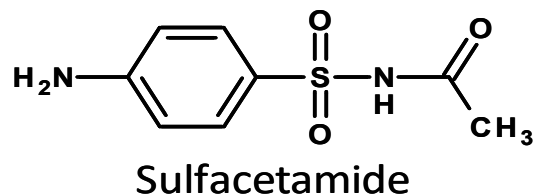
Therapeutic Application

Sulfacetamide

- Available under various brand names
- For treatment of eye infections

Silver sulfadiazine (Silvadene®)

- Silver salt only slightly soluble
- silver ions also provide antibacterial effect
- Used topically for treatment/prevention of infection in second and third degree burns
- Prevents the growth of fungi and bacteria including *Pseudomonas aeruginosa*



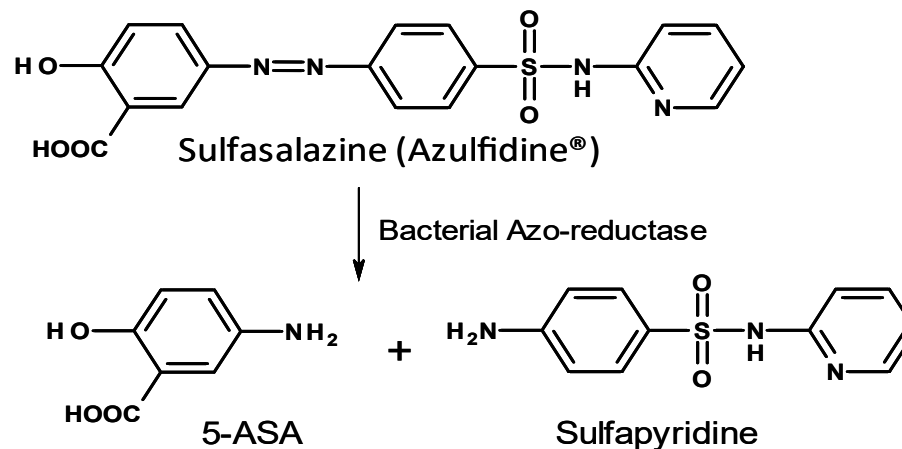
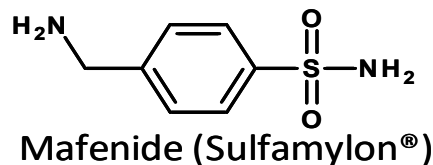
Therapeutic Application

Mafenide (Sulfamylon®)

- Used topically as an adjunct in the treatment of second and third degree burns to prevent septicemia

Sulfasalazine (Azulfidine®)

- Used for the treatment of inflammatory bowel disease
- metabolized by intestinal bacteria into its components, sulfapyridine and mesalamine (5-ASA)
- The anti-inflammatory effect is believed to be due to 5-ASA

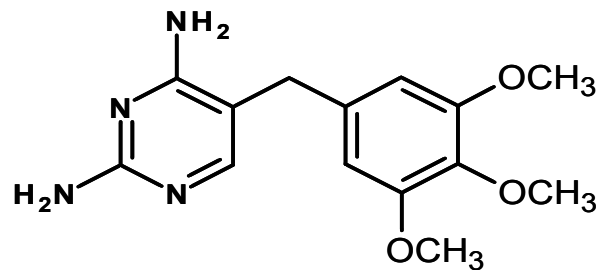


Adverse Effects of Sulfonamides

- Crystalluria – drugs may precipitate out as crystals and be deposited in the kidneys
- Recommended to drink increased quantities of water when taking certain sulfonamides to avoid crystalluria
- Hypersensitivity problems occur with about 5% of patients
- Hypersensitivity reactions include fever, rash, photosensitivity.

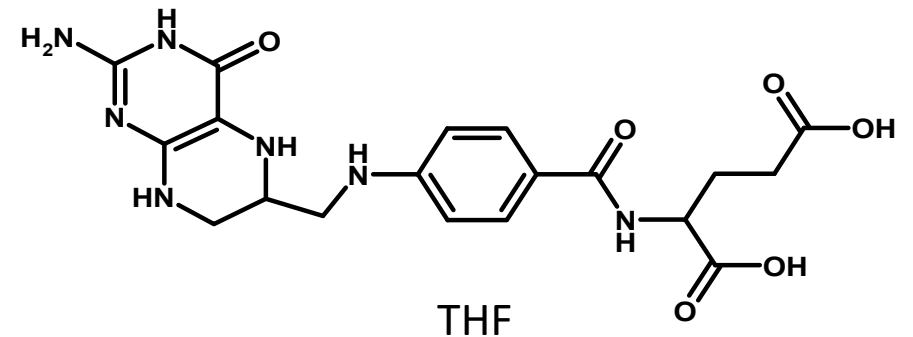
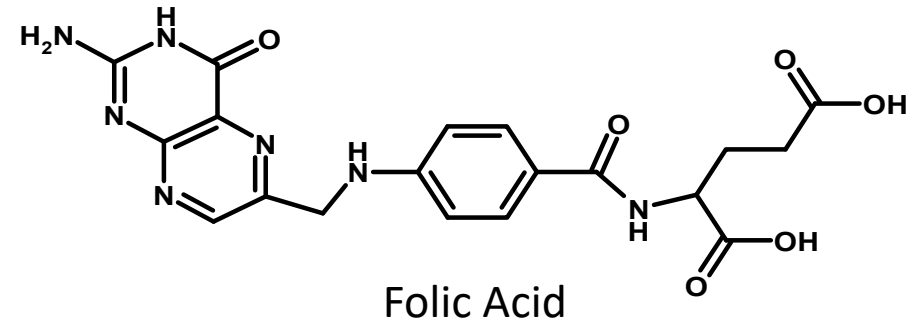
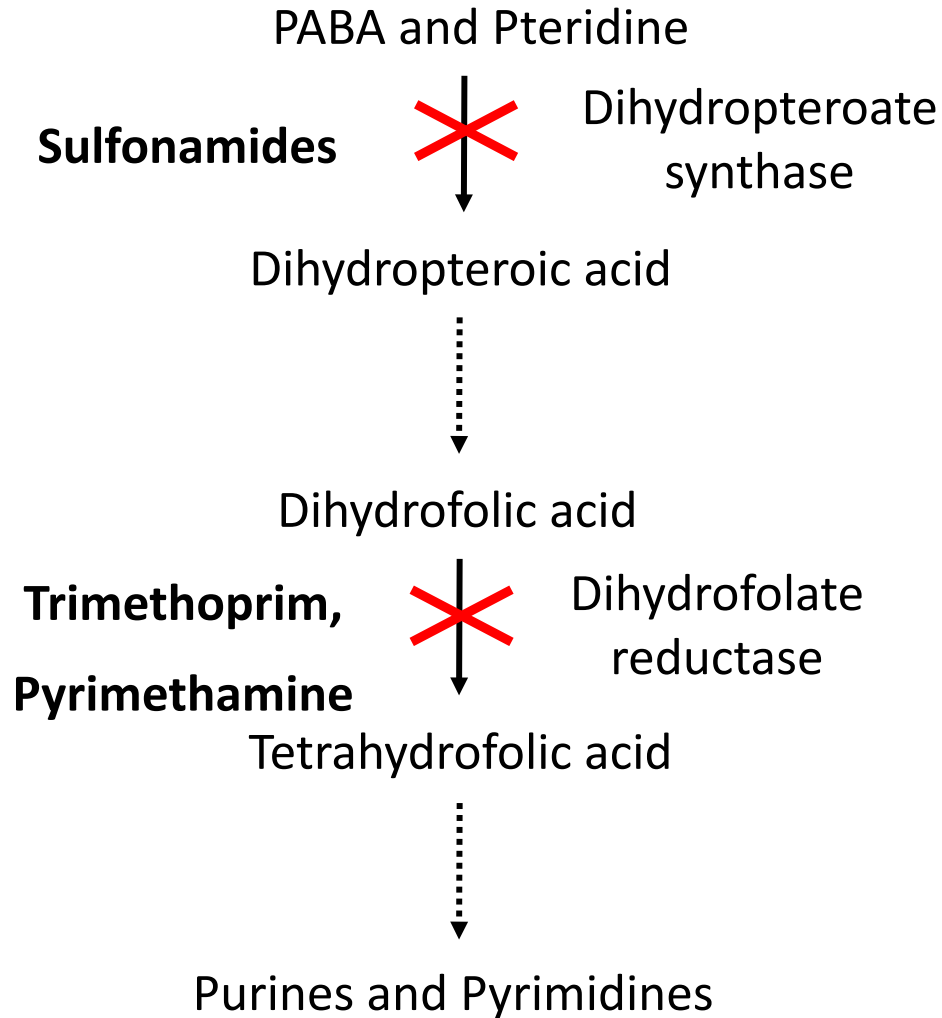
Other Antimetabolites: Trimethoprim

- Discovered in 1956 by George Hitchings and Gertrude Elion (marketed in 1969)
- Trimethoprim inhibits dihydrofolate reductase
- dihydrofolate reductase converts dihydrofolate to THF
- Dihydrofolate reductase is encoded by mammalian and bacterial cells
- Trimethoprim is 100,000 more active against bacterial dihydrofolate reductase



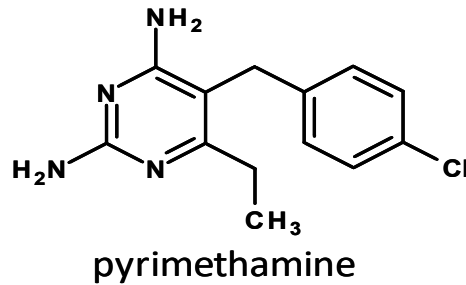
Trimethoprim

Mechanism of Action of Trimethoprim



Other Antimetabolites: Pyrimethamine

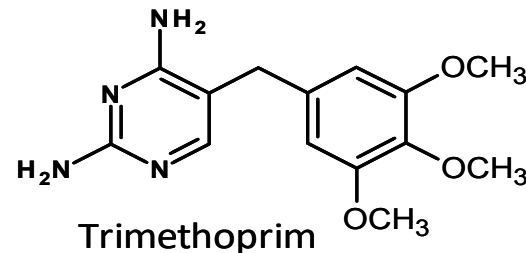
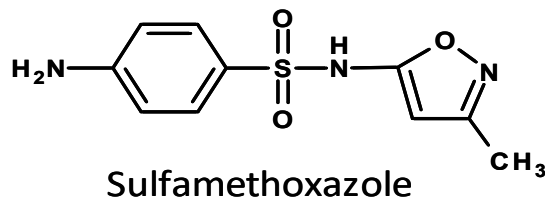
- Pyrimethamine inhibits malarial dihydrofolate reductase (DHFR)
- Has significantly higher binding affinity to the plasmodium DHFR than to the host enzyme ($> 1000X$ in *P. Berghei*)



Therapeutic Application

Sulfamethoxazole with trimethoprim (Bactrim®, Septra®)

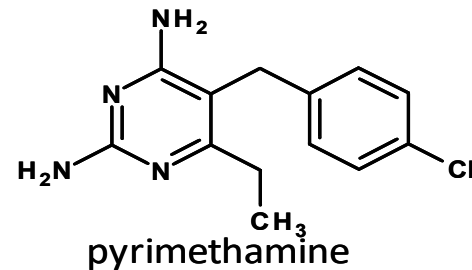
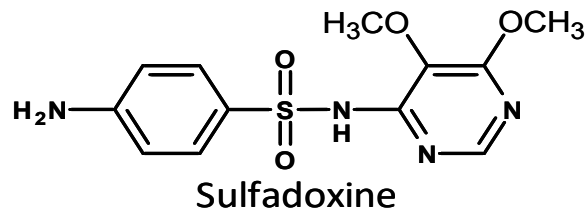
- Inhibit at two different steps in the same biosynthetic pathway
- Have synergistic effect
- For the treatment of UTI caused by enterobacteriaceae including *E. coli* and *Klebsiella*
- Otitis media - *H. influenzae*, *S. pneumoniae*
- *Pneumocystis carinii* pneumonitis (PCP) prophylaxis and traveler's diarrhea



Therapeutic Application

Sulfadoxine with pyrimethamine (Fansidar®)

- Used for the treatment of *P. falciparum* malaria when chloroquine resistance is suspected
- For malaria prophylaxis for travelers to areas where chloroquine-resistant malaria is endemic
- Sulfadoxine is an ultra-long-acting sulfonamide; half-life 4-9 days (only taken once/week)

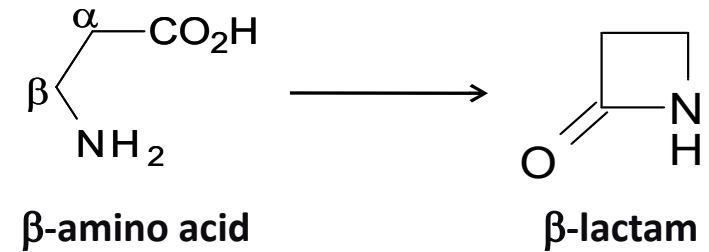


2. Bacterial Cell Wall Synthesis Inhibitors

A. β -lactams

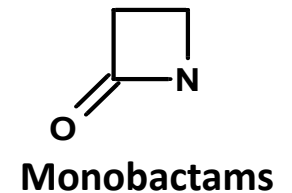
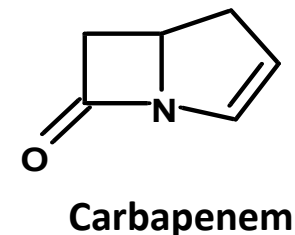
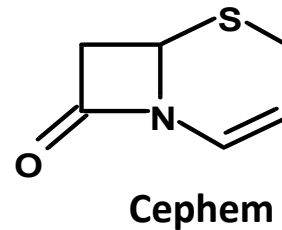
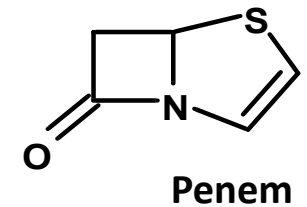
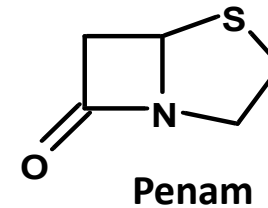
- Penicillins (penams)
- Cephalosporins (Cephems)
- Penems
- Carbapenems
- Monobactams

Lactam = Cyclic amide



B. Glycopeptides

- Vancomycin



Bacterial Cell Wall Synthesis Inhibitors

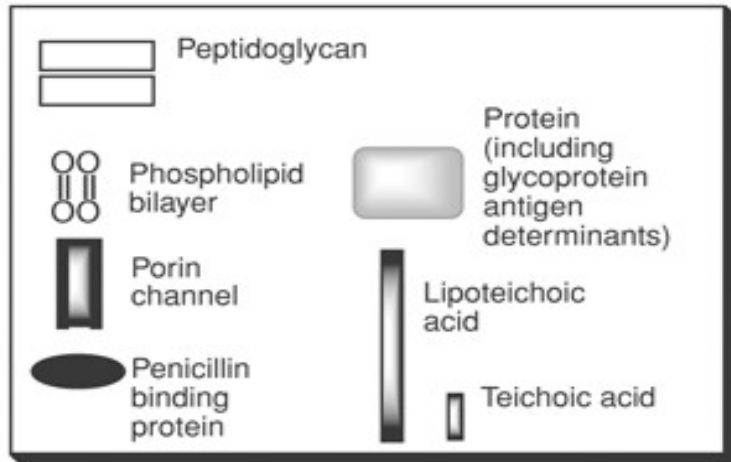
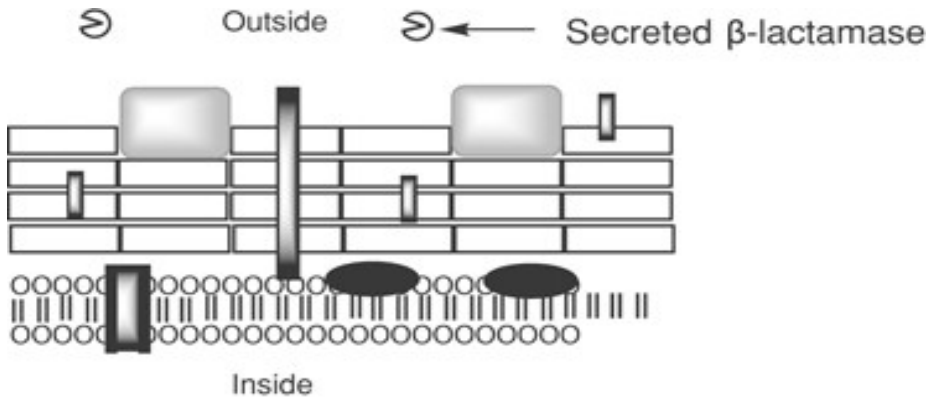
Bacterial cell wall

- Bacterial cells enclosed within a complex and rigid cell wall
- Mammalian cells are surrounded by a flexible membrane, the chemical composition of which is dramatically different from that of the cell wall
- Enzymes that are involved in the synthesis of bacterial cell wall do not have mammalian counterpart
- Therefore, attractive targets for selective chemotherapy of bacterial infections

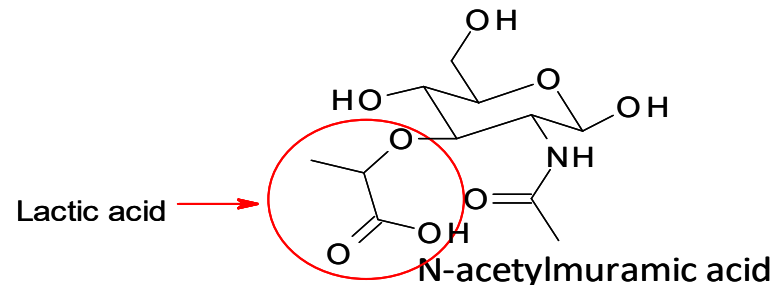
Functions of cell wall

- Provides a selective barrier interfacing with the environment through which only desirable substances may pass
- Maintains osmotic balance
- prevents digestion by host enzymes

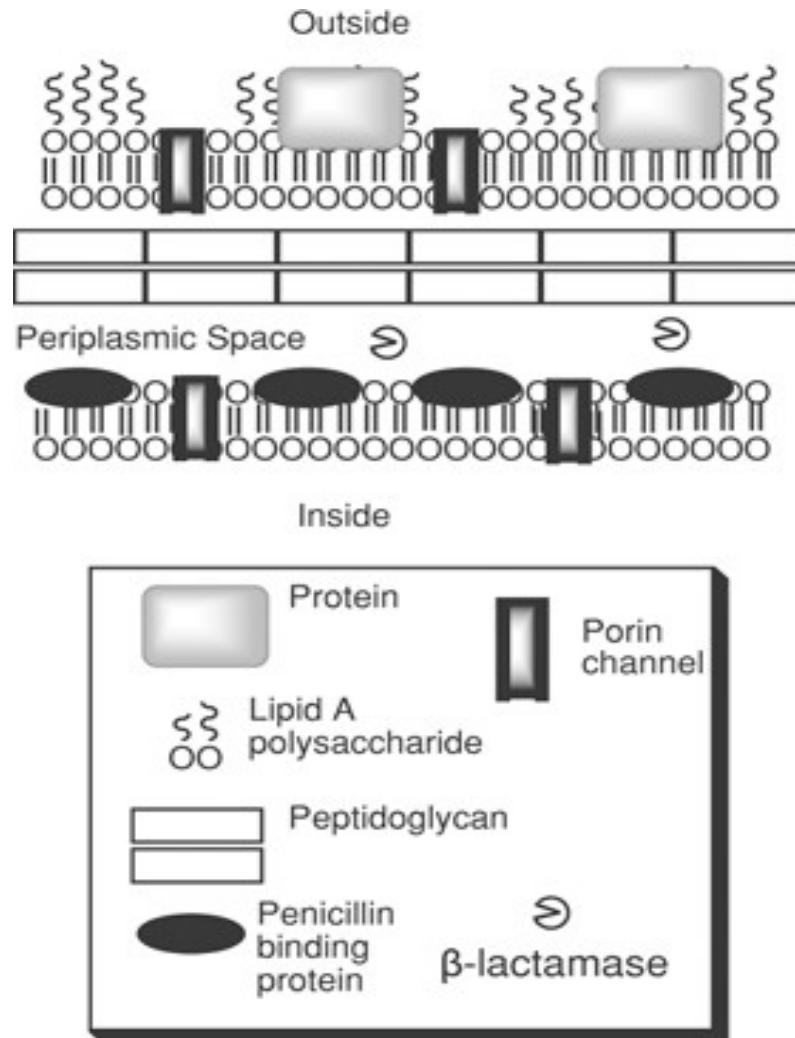
Gram-Positive Bacterial Cell Wall



- Complex enough but simpler than that of Gram-negative organisms
- The peptidoglycan layer is a spongy, gel-forming layer consisting of a series of alternating sugars (N-acetylglucosamine and N-acetylmuramic acid) linked (1,4)- β in a long chain
- A series of amino acids (e.g. L-alanyl-D-glutamyl-L-lysyl-D-alanine in *S. aureus*) is attached to the lactic acid carboxyl moieties of the N-acetylmuramic acid units through an amide linkage
- The D-stereochemistry of the Glu and Ala makes these peptides unlikely substrate of host peptidases

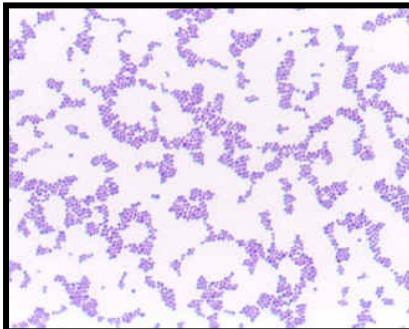
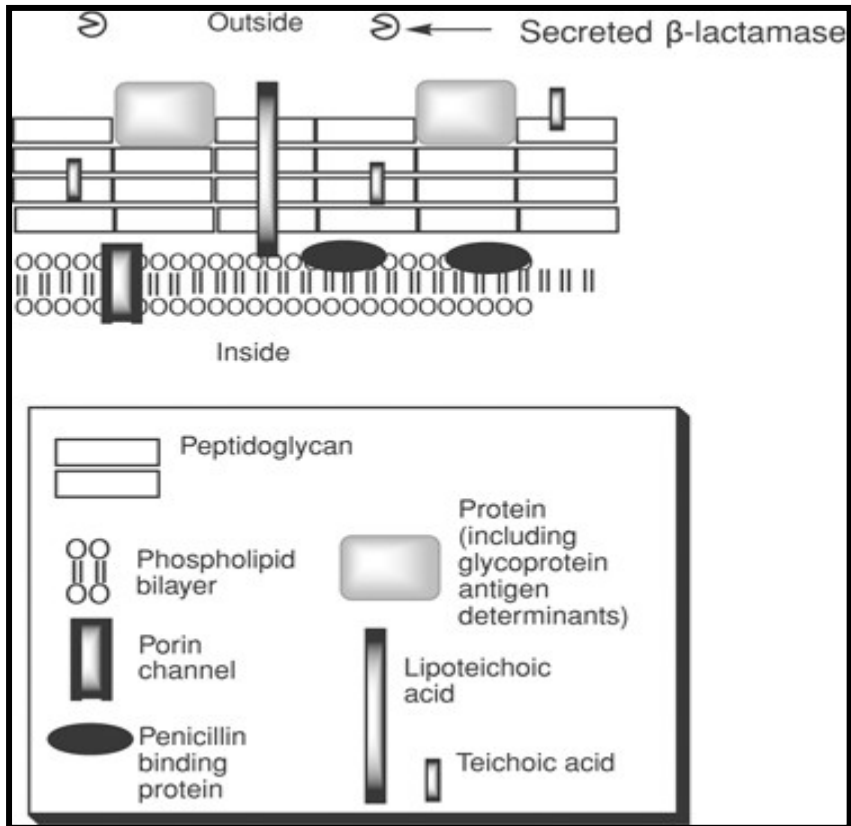


Gram-Negative Bacterial Cell Wall

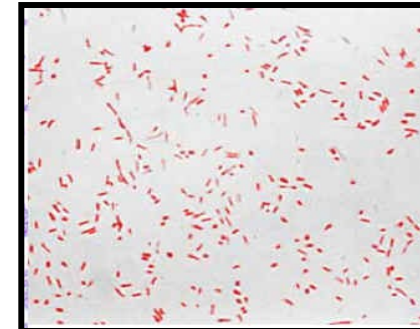
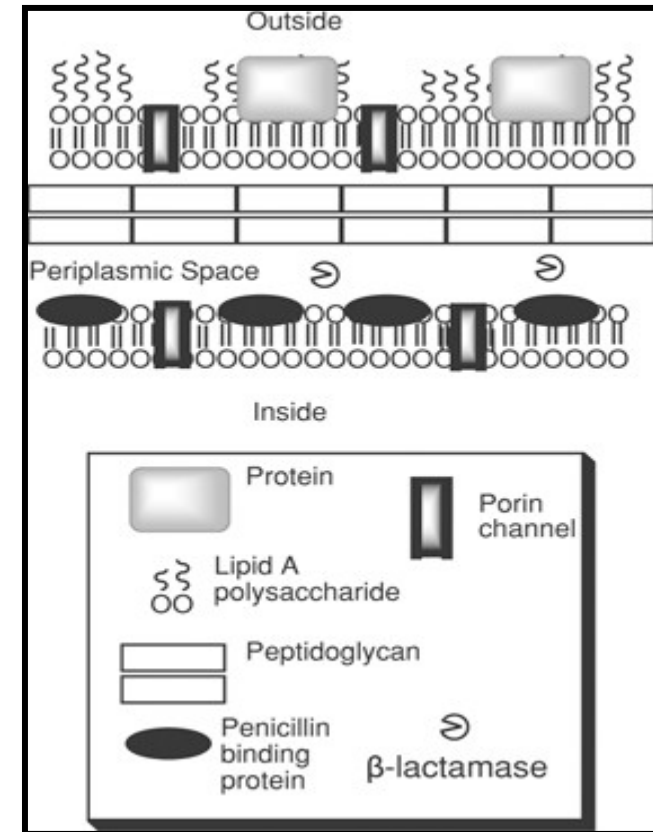


- The cell wall is more complex and has additional outer lipid membrane that differs significantly from the inner membrane
- The outer layer contains complex lipopolysaccharides that encode antigenic responses, cause septic shock, provide the serotype, and influence morphology
- The outer layer also contains a number of enzymes and other proteins including porins
- Porins are transmembrane molecules made up of two or three monomeric proteins
- The size, shape, and lipophilicity of drugs are important considerations controlling porin passage
- Antibiotics have greater difficulty in penetrating into Gram-negative bacterial cells as a result

Gram-positive bacteria cell wall

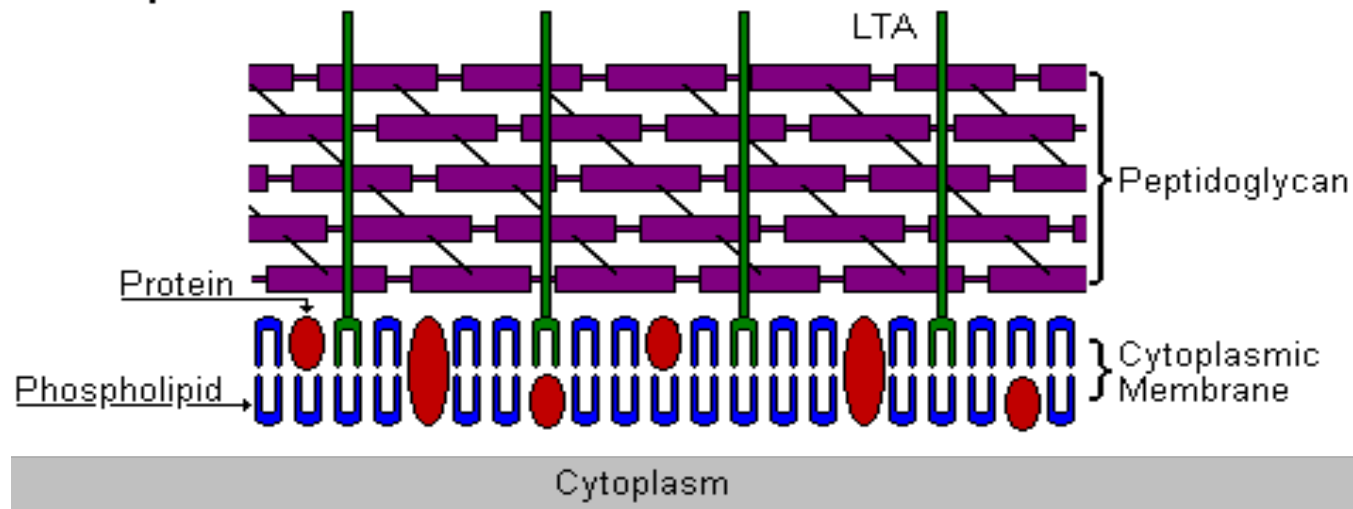


Gram-negative bacteria cell wall

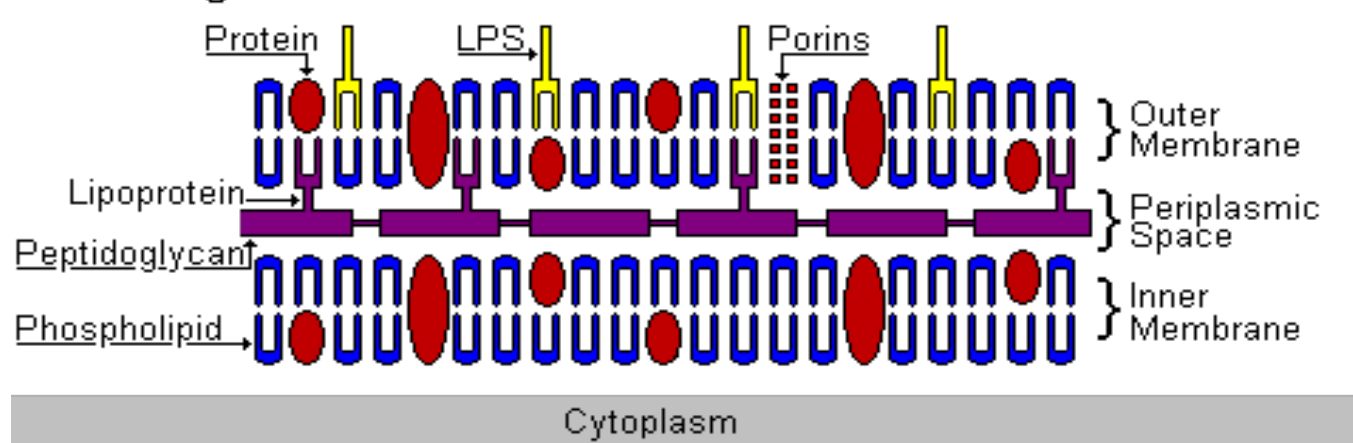


Cell Wall Structure of Gram(+) and Gram(-) Bacterium

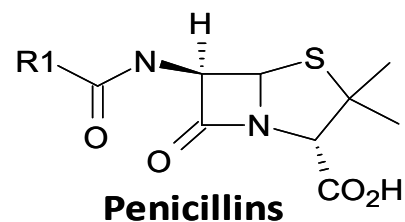
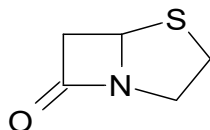
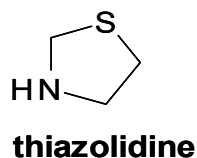
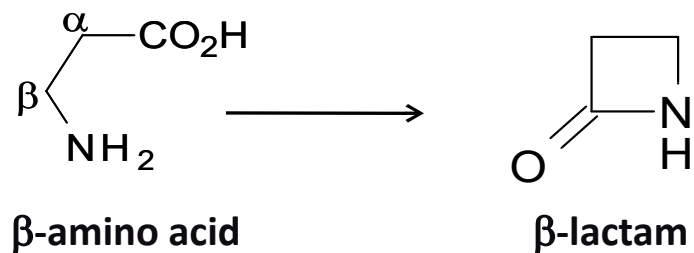
Gram-positive Cell Wall



Gram-negative Cell Wall



β -Lactam Antibiotics



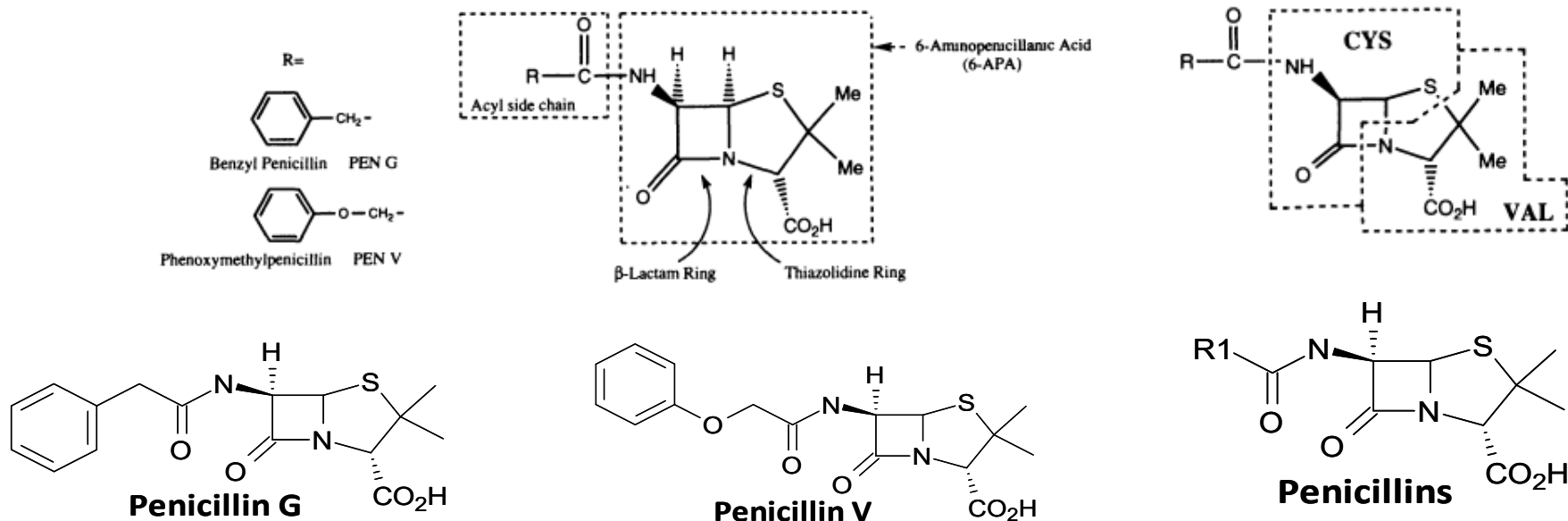
- A lactam is a cyclic amide
- The β -lactam ring is the main component of the pharmacophore
- In the penicillin subclass of β -lactam antibiotics, a substituted 5-membered thiazolidine ring is fused to the β -lactam ring

History of Penicillins

- 1929- Penicillin was discovered accidentally by Alexander Fleming
- A. Fleming left a petridish containing *Staphylococcus* open
- The culture of the bacteria was contaminated by mould
- Fleming observed that bacterial growth was inhibited around the mould, leading to his conclusion that the inhibition was due to a substance released by the mould
- A pure culture of the mould was grown and later named *Penicillium notatum*
- The name penicillin was given to the filtrate of a broth culture of the *Penicillium* mould
- This filtrate was found to contain a mixture of up to five compounds that were designated as penicillin G, X, F, K
- The yield was improved by using a more productive species, *Penicillium chrysogenum* and by addition of metal salts and side-chain precursors

History of Penicillins

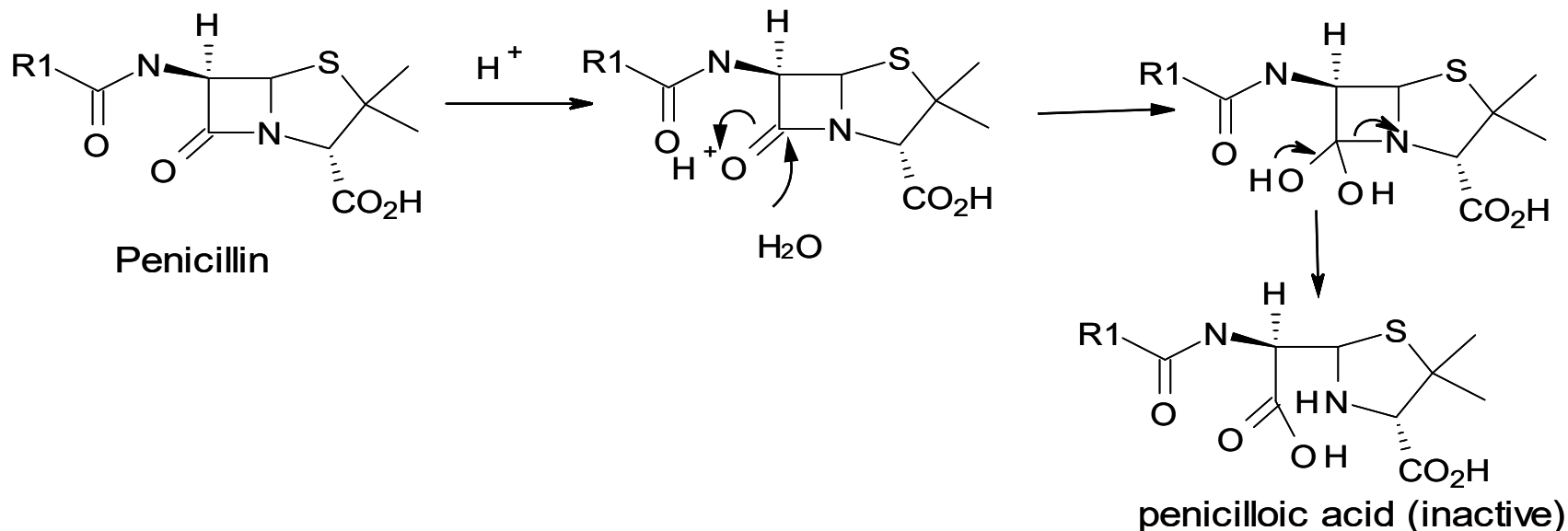
- Penicillin G (benzyl penicillin) was chosen due to its high effectiveness, and the ability to obtain high yields by addition of phenylacetic acid to the culture
- Penicillin G also had low serum binding property compared to the other penicillins
- However, penicillin G was found to be unstable in the acidic environment of the stomach
- This led to an effort to produce new penicillins through addition of other types of precursor acids ($R_1\text{COOH}$)
- The effort resulted in the production of several new penicillins, among which Penicillin V (Phenoxymethyl penicillin) is still in use today



Ring opening of the β -lactam ring under acidic conditions

Reasons for acid instability

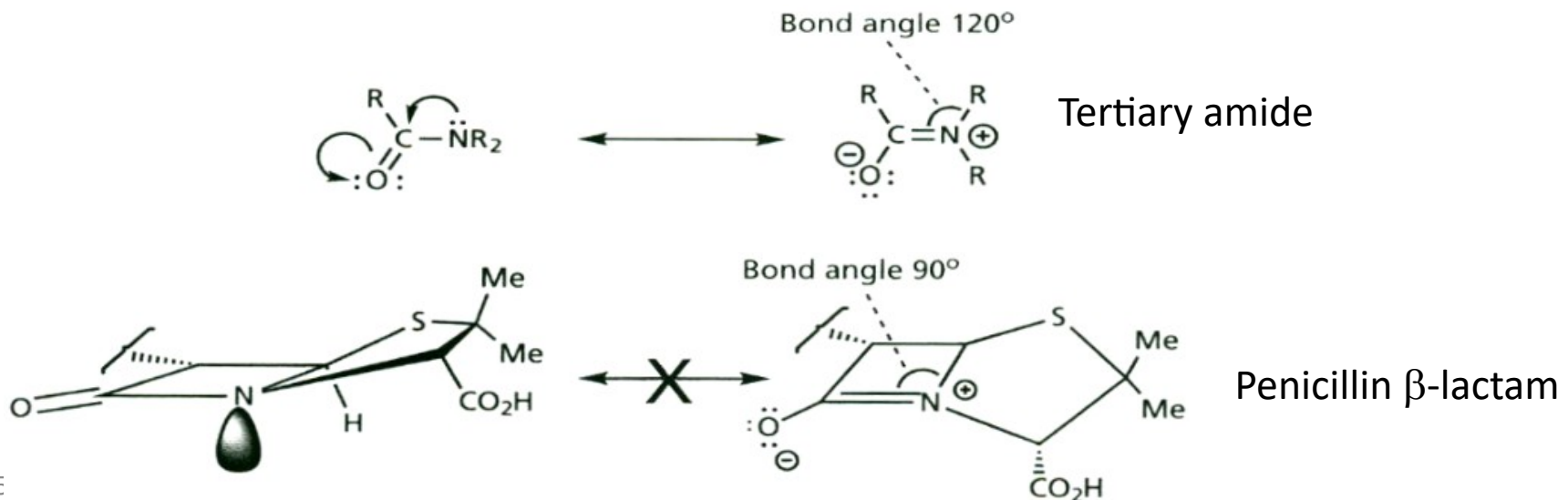
1. Ring strain: bicyclic ring system induces large angle and torsional strains in penicillins. Acid catalyzed ring opening relieves these strains by breaking open the more highly strained β -lactam ring
- The β -lactam ring is an essential portion of the pharmacophore and its hydrolysis deactivates the antibiotic



Ring opening of the β -lactam ring under acidic conditions

Reasons for acid instability

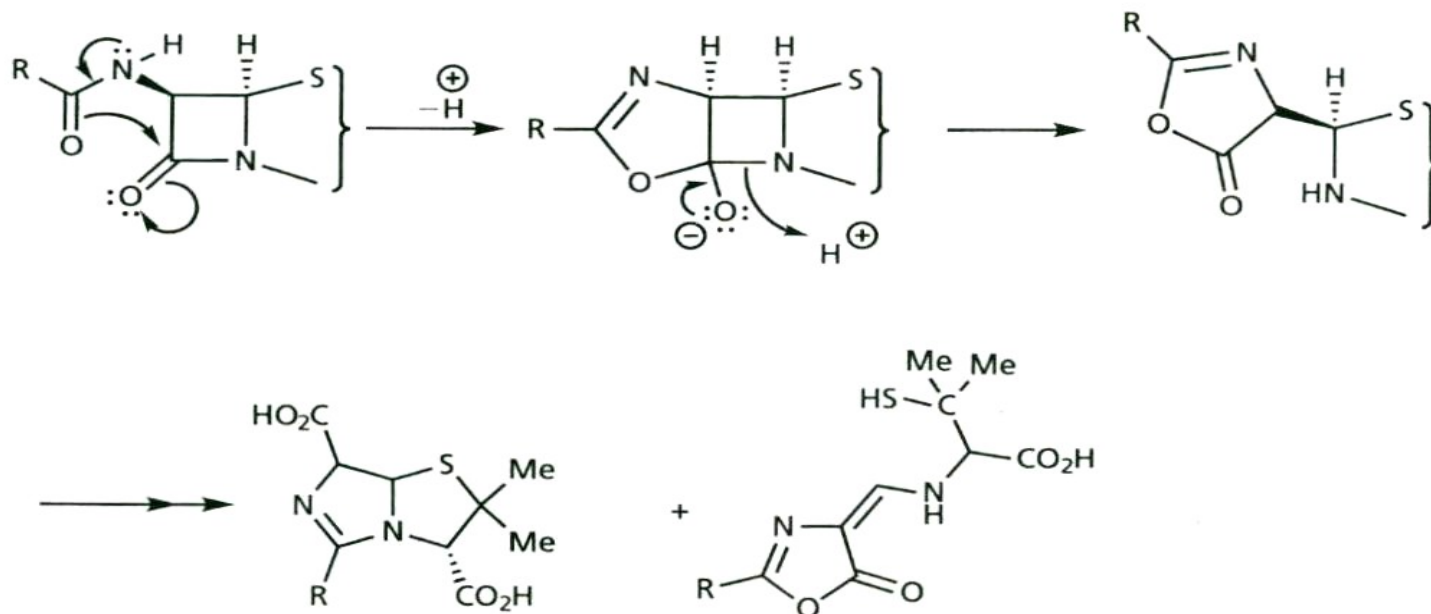
2. A highly reactive β -lactam carbonyl group: the carbonyl group in the β -lactam ring is highly susceptible to nucleophiles and does not behave like a normal tertiary amide
- A normal tertiary amide is resistant to nucleophilic attack as the carbonyl group is stabilized by neighboring nitrogen atom through resonance
 - Resonance stabilization is impossible for the β -lactam ring, because the increase in angle strain that would result from having a double bond within a 4-membered β -lactam ring
 - As a result, the lone pair of electrons is localized on the nitrogen atom, and the carbonyl group is more electrophilic than one would expect for a tertiary amide.



Ring opening of the β -lactam ring under acidic conditions

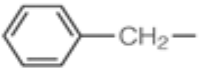
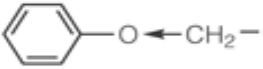
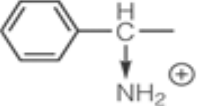
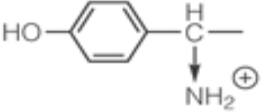
Reasons for acid instability

3. Influence of the acyl side chain at the 6-position (neighboring group participation): the acyl side chain at the 6-position can actively participate in a mechanism to open up the β -lactam ring. Thus, penicillin G has a self destructive mechanism built into it

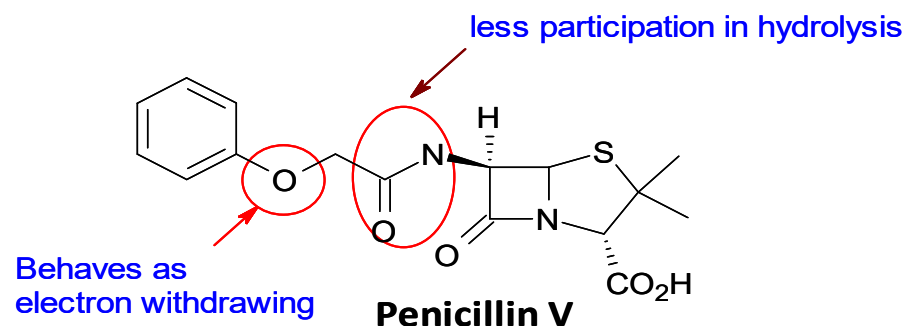


Influence of the acyl side chain at the 6-position on the acid sensitivity of penicillins

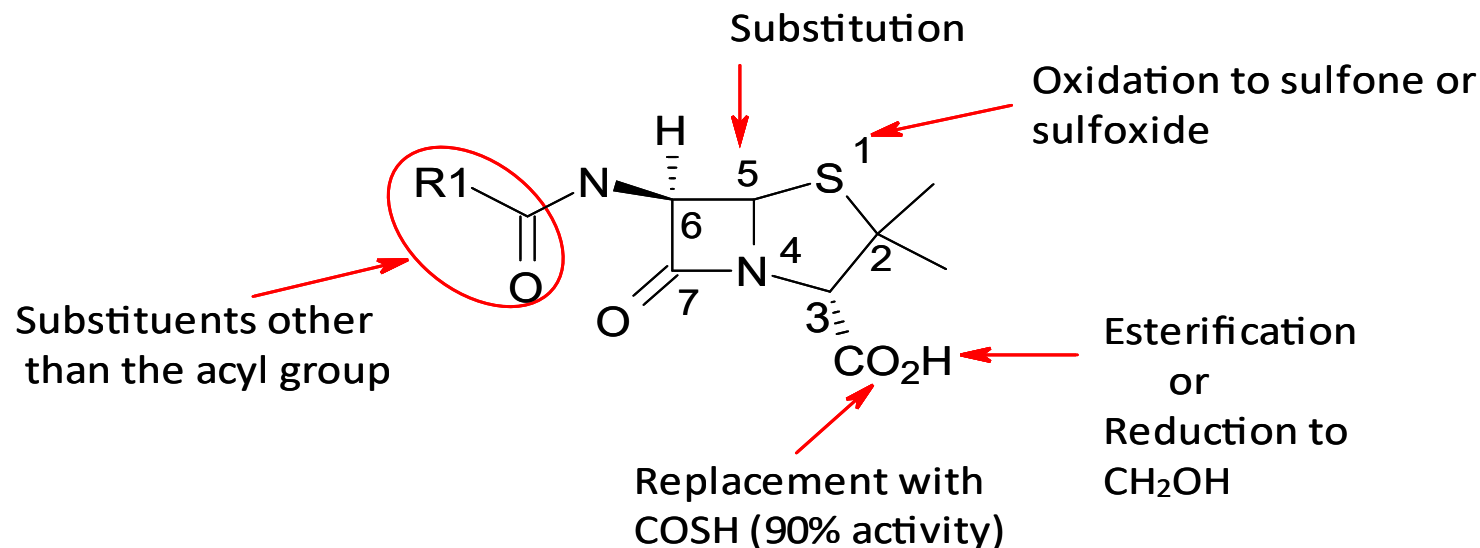
Improved Acid Stability and Absorption of Substituted Penicillins

Drug	R	% Absorption intact drug
Benzylpenicillin		15–30
Pen V		60–73
Ampicillin		30–50
Amoxicillin		75–90

- Substitution of a side-chain R with an electron-withdrawing group (EWG) decreases the electron density on the side-chain carbonyl and therefore decreases the side chain participation
- Because of their acid stability, compounds with EWG at the side chain including Penicillin V, ampicillin and amoxicillin are given orally

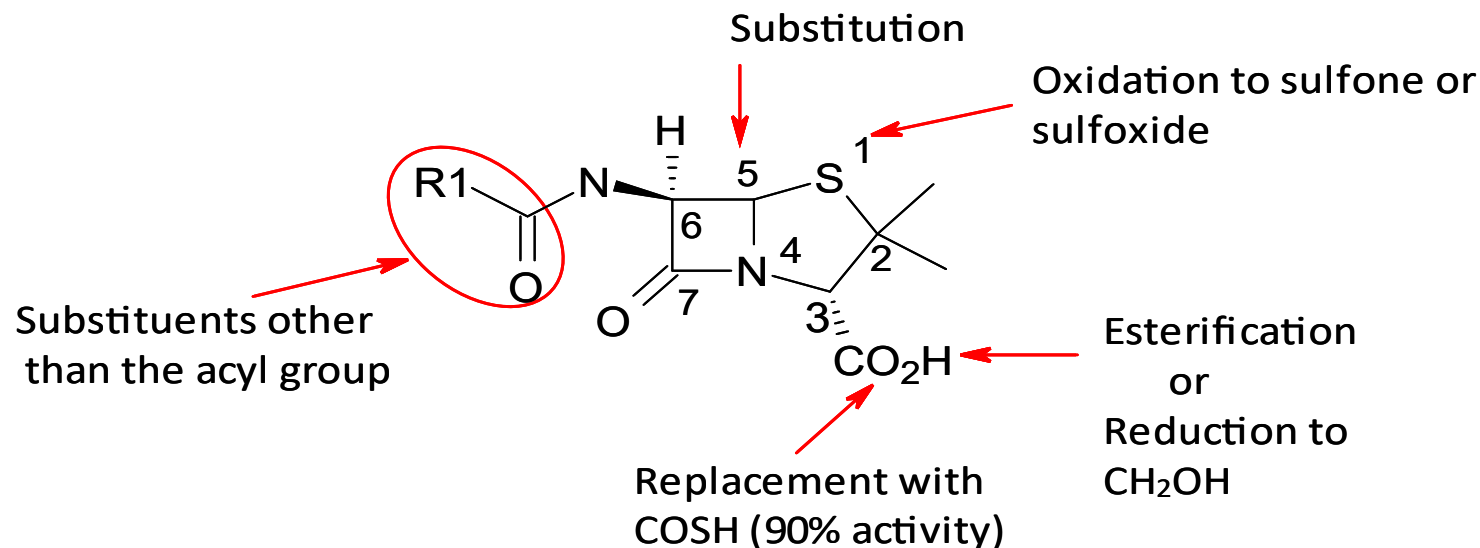


Structure-Activity Relationship (SAR) of Penicillins



- Efforts to modify several parts of the penicillin structure (other than the acyl moiety) resulted in either in dramatic reduction or loss of activity
- Oxidation of the sulfur atom to sulfoxide or sulfone increased the acid stability of the compound but decreased the antibacterial activity significantly
- Any substitution at the 5-position of the ring system resulted in inactive compounds
- Esterification (e.g. methyl ester) or reduction to hydroxymethyl group of the 3-carboxylic moiety resulted in loss of antibacterial activity

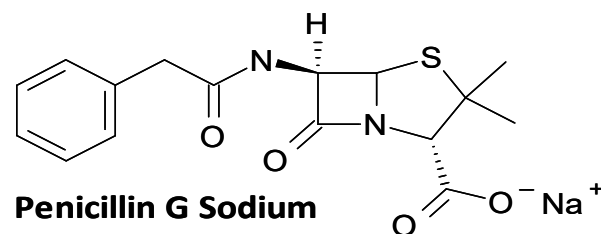
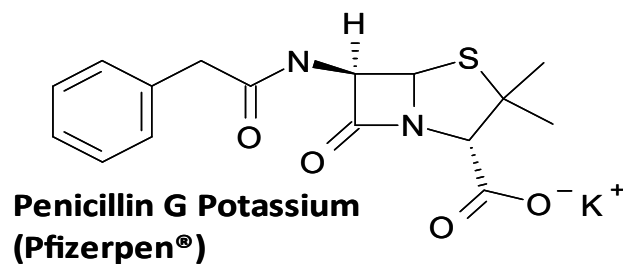
SAR of Penicillins (continued)



- Conversion of the 3-carboxyl group to carboxamide (CONH₂) retains only 25% of the activity of the parent compound
- Only replacement of 3-carboxylic group with the thio isostere (-COSH) retains significant amount of activity (90% of the activity)
- All clinically useful penicillins vary only at the R1 position (have different R1 substituents)

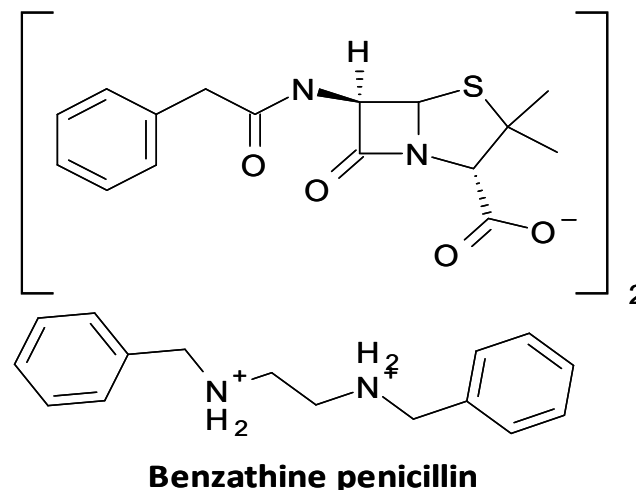
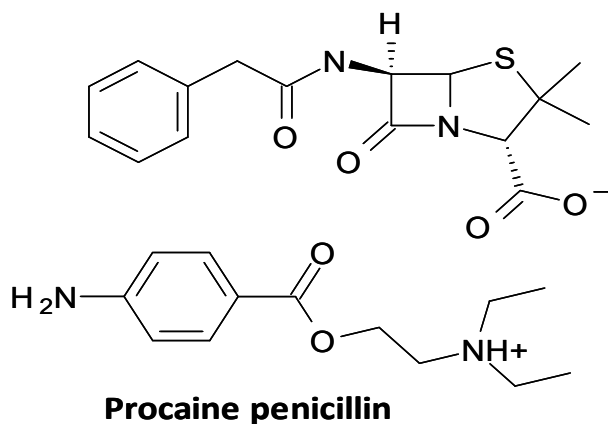
SAR of Penicillins: penicillin G

- The potassium and sodium salts of Penicillin G are water soluble and are given parentally
- Penicillin G potassium and penicillin G sodium are available as powder for reconstitution
- Due to its efficacy, low cost, and lack of toxicity (except for acutely allergic patients), Penicillin G remains an extremely useful agent for the treatment of infections caused by susceptible microorganisms
- Except for *Neisseria gonorrhea* and *Haemophilus influenza*, the primary use of penicillin G is limited to infections caused by Gram-positive cocci



SAR of Penicillins: Penicillin G

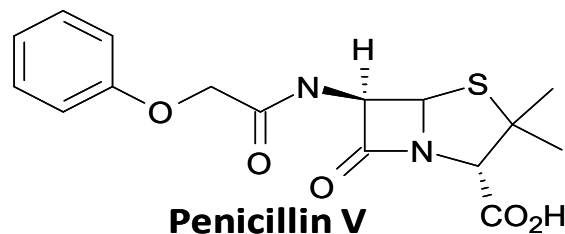
- Benzathine penicillin and Procaine penicillin are salts of Penicillin G with organic bases benzathine and procaine, respectively
- These are sparingly soluble in water and used for deep intramuscular (IM) injections
- Produce lower but prolonged blood levels of penicillin G as the drug slowly diffuses from the injection site
- Due to low blood concentrations produced, their use should be restricted to infections caused by micro-organisms that are highly sensitive to penicillin G
- Should not be used as the sole treatment for severe acute infections, or when bacteraemia is present



SAR of Penicillins

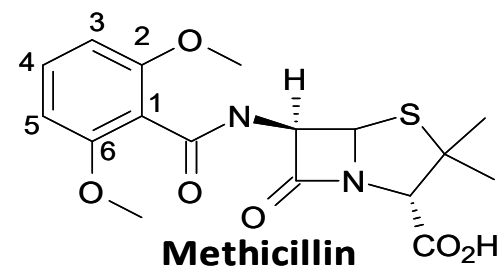
Penicillin V

- is acid-stable and can be given orally
- Its antibacterial spectrum and potency is similar to that of penicillin G
- Two problems remains to be addressed are: sensitivity to β -lactamases (resistance) and the narrow antimicrobial spectrum of activity



Penicillinase-Resistant Parenteral Penicillins

- Methicillin was the first penicillinase-resistant penicillin in clinical use
- The rate of hydrolysis of methicillin by staphylococcal β -lactamase was 1% of that of penicillin G
- However, **methicillin is not acid stable**
- Methicillin has a narrower antimicrobial spectrum and less potency
- Therefore, methicillin was restricted to clinical use mainly for parenteral use in infections caused by **β -lactamase-producing *Staphylococcus aureus*** and a few other infections



Methicillin Resistance

- Infections caused by methicillin-resistance *Staphylococcus aureus* (MRSA) have been on the rise

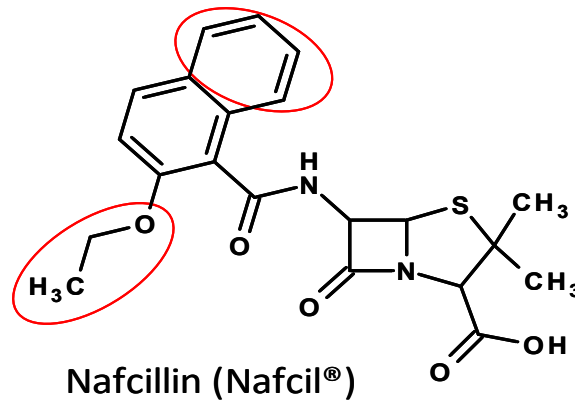
The mechanism of resistance in these cultures seems to be due to

- **reduced uptake** of the drug a
- **alteration in the penicillin binding proteins (PBPs)**
- Especially, an altered PBP-2 with very low affinity for β -lactams including methicillin is formed
- In addition, methicillin is an **efficient inducer of penicillinases**
- As a result this drug fell out of favor and it has now been replaced by a number of agents

Penicillinase-Resistant Parenteral Penicillins

Nafcillin (Nafcil®)

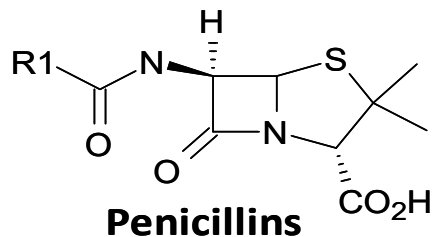
- Clinically similar to methicillin clinically but slightly more acid-stable



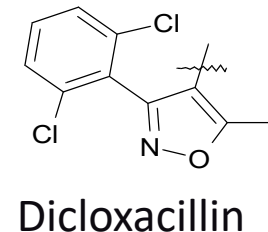
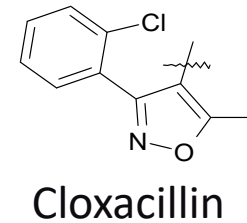
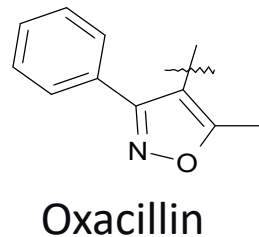
Penicillinase-Resistant Parenteral Penicillins

Oxacillin, cloxacillin, and dicloxacillin

- have an **isoxazolyl ring** as a **bio-isosteric** replacement for the **benzene ring**
- The dimethoxy groups of methicillin are also replaced by methyl and substituted or unsubstituted benzene ring
- differ from one another by the substituents on the benzene ring
- **less potent than penicillin G** against Gram-positive microorganisms that **do not produce** a β -lactamase but retains their activity against those G⁺ bacteria that produce β -lactamase
- Slightly more acid resistant than penicillin G and may be taken orally
- Due to their high serum protein binding, they are not good choices for treatment of septicemia (Invasion of the bloodstream by virulent microorganisms from a focus of infection).
- Usually microorganisms resistant against methicillin also are resistant to the Oxacillin, cloxacillin, and dicloxacillin

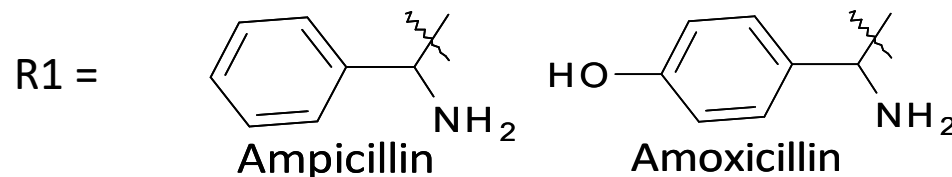
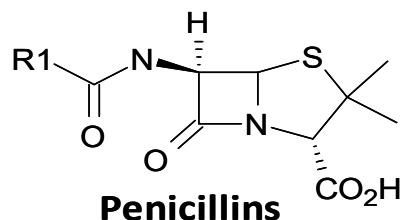


R1 =



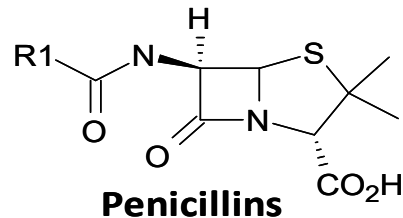
Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins

- **Ampicillin** and **amoxicillin** are benzylpenicillin analogues in which one of the hydrogen atoms of the side-chain methylene has been replaced with a primary amino group
- Introduction of an amine group at the benzyl position rendered the molecules to be stable in acidic conditions
- This modification has also resulted in broad-spectrum activity (G- activity) i.e. the antimicrobial spectrum is shifted so that many common Gram-negative pathogens are sensitive to ampicillin and amoxicillin
- **The broad spectrum property is believed to be due to greater penetration of ampicillin and amoxicillin into Gram-negative bacteria**

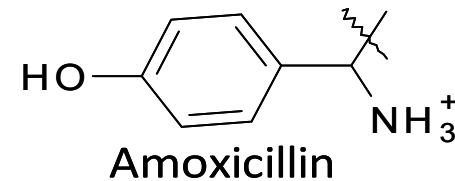
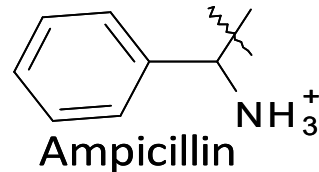


Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins

- The acid stability of ampicillin and amoxicillin is thought to be due to the electron withdrawing character of the protonated primary amine group, reducing participation in hydrolysis of the β -lactam
- Ampicillin is partly absorbed by the small intestine peptide carrier, increasing its oral activity
- Although ampicillin and amoxicillin are acid-stable, they lack stability toward β -lactamases
- To address the problem of lack of resistance towards β -lactamase both are given with β -lactamase inhibitors (will be discussed later)

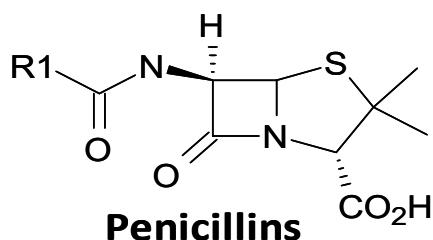


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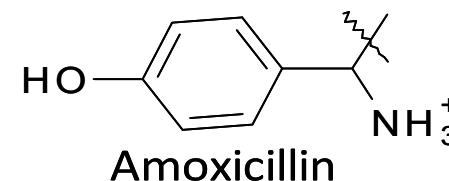
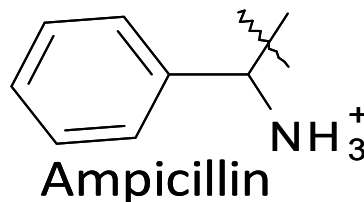


Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins

- Introduction of a para-hydroxyl group into the side-chain phenyl moiety increases the solubility of amoxicillin in the gut leading to increased oral absorption (74–92%)
- As a result amoxicillin causes less disturbance of the normal GI flora and, therefore, less drug-induced diarrhea compared to ampicillin

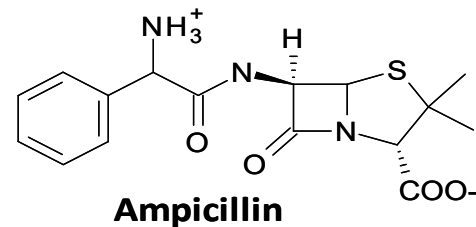


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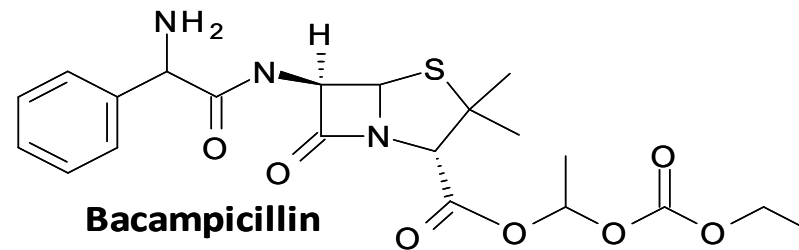


Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins

- The oral bioavailability of ampicillin can be enhanced significantly by the synthesis of pro-drugs that have better absorption
- Bacampicillin, a prodrug of ampicillin is very well absorbed in the duodenum (80–98%)
- Hydrolysis of bacampicillin by esterases gives carbon dioxide and ethanol, followed by spontaneous loss of acetaldehyde and production of ampicillin

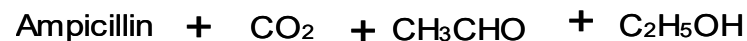


amphoteric, 30-55% absorption



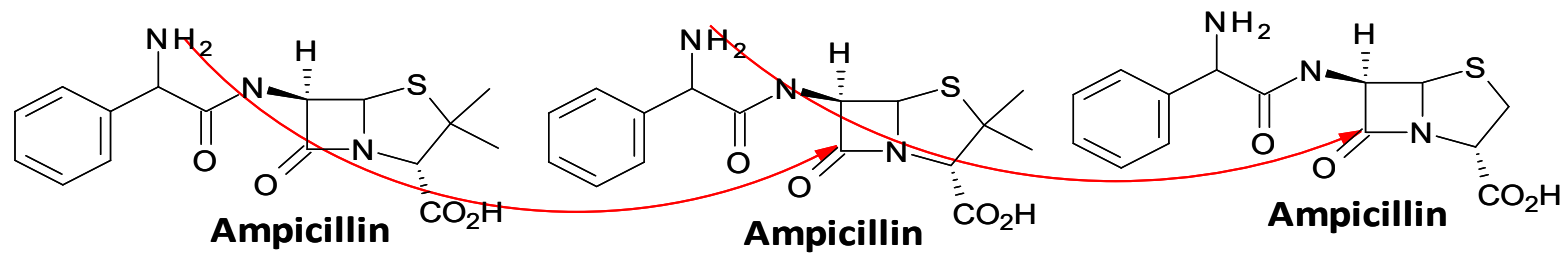
weak base, oral absorption 80-98%

hydrolysis



Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins

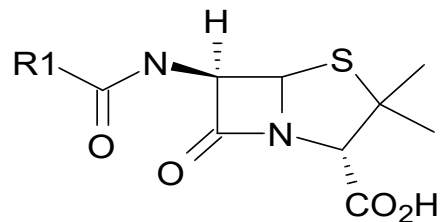
- In addition to the common mode of penicillin allergenicity, concentrated preparations of ampicillin can self-condense to form high-molecular-weight aggregates through reaction of its primary amino group with the β -lactam bond of another molecule
- These aggregates are thought to be antigenic and to be responsible for ampicillin allergenicity
- Ampicillin and amoxicillin are the penicillins most commonly associated with drug-induced rash



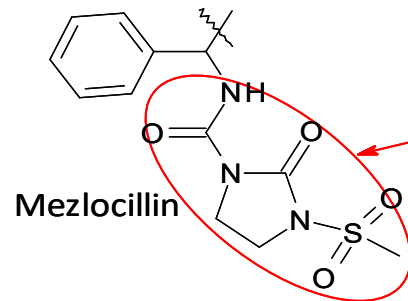
Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins

Mezlocillin and piperacillin

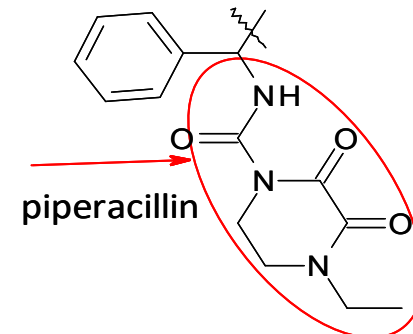
- are known as acylureidopenicillins
- Introduction of a ureidyl moiety at the benzylic position extended the spectrum of activity further
- Active against some strains of *Pseudomonas aeruginosa*



R1 =



mimics a longer
segment of the
peptidoglycan



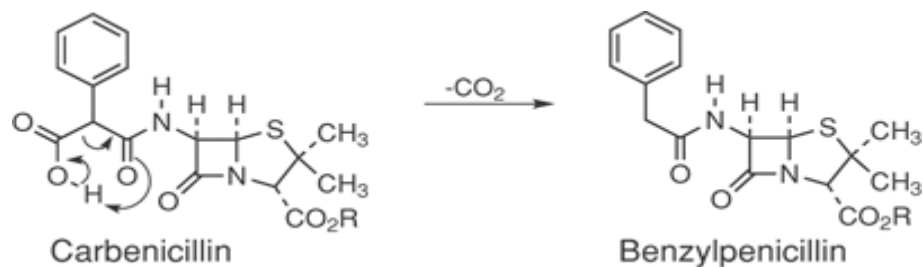
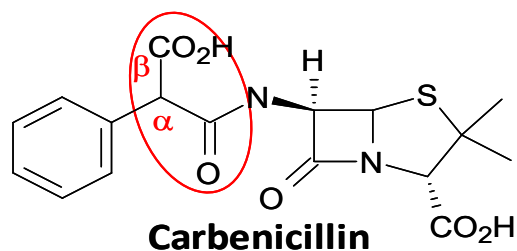
Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins

Mezlocillin and piperacillin

- acid labile and not resistant to β -lactamases
- used parenterally for the treatment of infections caused by Gram-negative bacteria, especially *Klebsiella pneumoniae* and the anaerobe *Bacteroides fragilis*
- Addition of other antibacterial agent e.g. an aminoglycoside for the treatment of severe infections is advisable
- Piperacillin is available with a β -lactamase inhibitor tazobactam as Zosyn

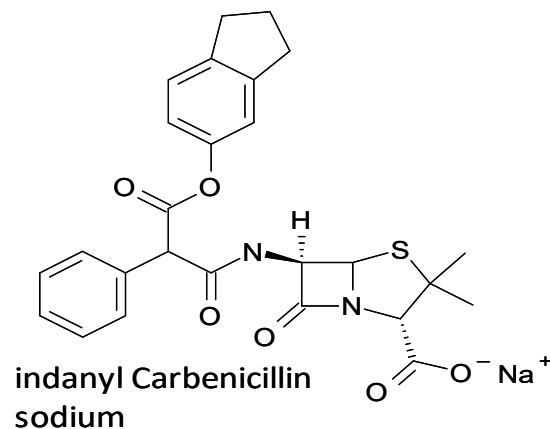
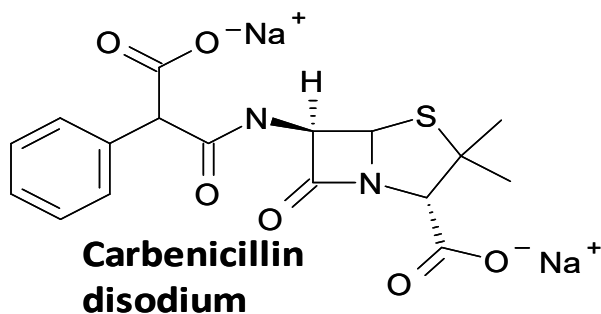
Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins

- Carbenicillin and indanyl carbenicillin
- Introduction of a carboxylic acid moiety at the benzylic position of penicillin G yields Carbenicillin
- The introduction of the side-chain carboxyl produces enhanced anti-Gram-negative activity
- Carbenicillin is 10 times less potent than the acylureidopenicillins (such as Mezlocillin and piperacillin)
- The drug is susceptible to β -lactamases and is unstable in acid, therefore should be given by injection
- carbenicillin can decarboxylate readily to produce benzylpenicillin because the carboxyl group is β - to the carbonyl group of the amide



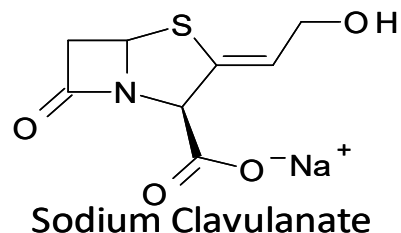
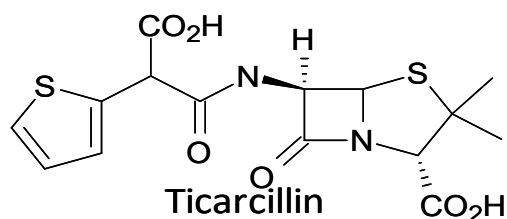
Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins

- Because of its low potency, a high dose (multi gram/day) of carbenicillin disodium is administered daily
- this large doses of carbenicillin sodium that results in ingestion of a significant amount of sodium ion, which could be a problem with heart patients
- Most of the problems were avoided by preparing indanyl carbenicillin, an oral ester prodrug
- Indanyl carbenicillin is mainly used orally for the treatment of urinary tract infections



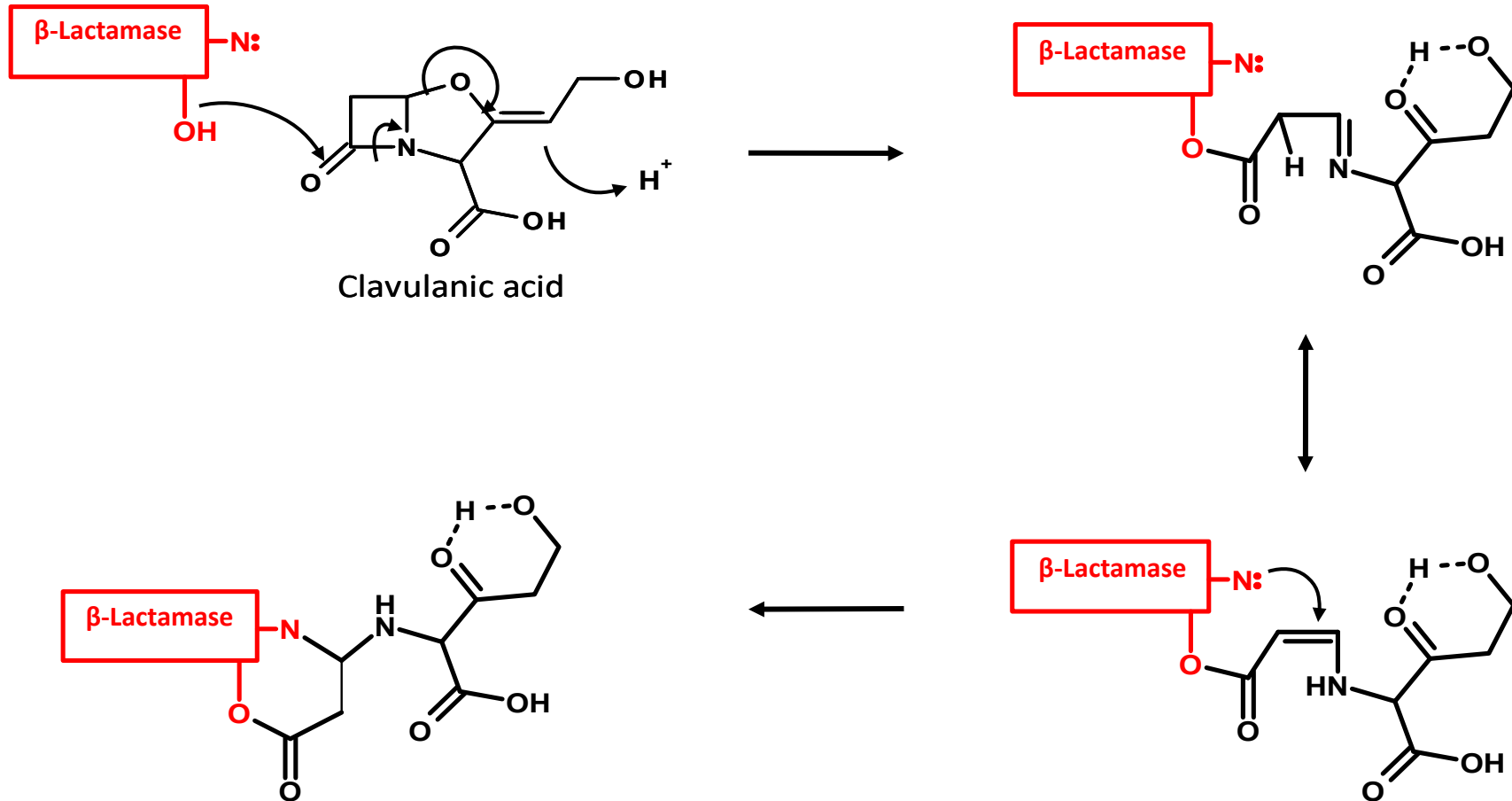
Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins

- Replacement of the phenyl group in carbenicillin with a thiophene bioisostere gave Ticarcillin
- Ticarcillin does not undergo decarboxylation like carbenicillin does
- Ticarcillin more potent than indanyl carbenicillin against pseudomonads
- When combined with the β -lactamase inhibitor potassium clavulanate, ticarcillin has enhanced antipseudomonas activity because of its enhanced resistance to β -lactamases

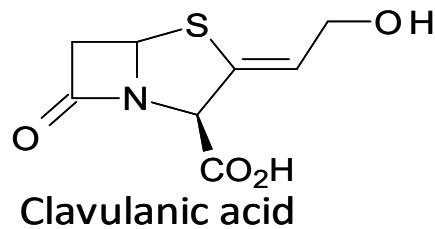


—————→ Timentin®

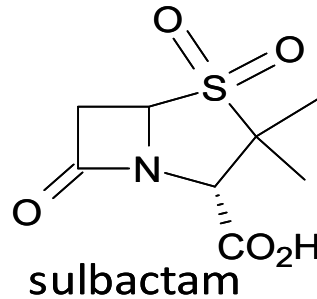
β -Lactamase Inhibitors: Mechanism of Action



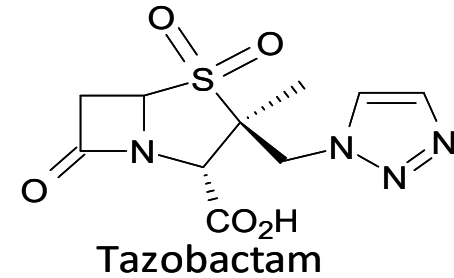
β -Lactamase Inhibitors



+
Amoxicillin
↓
Augmentin



+
Ampicillin
↓
Unasyn



+
Piperacillin
↓
Zosyn

- Efforts in the design and synthesis of β -lactamase inhibitors has resulted in several Clinically useful compounds

Penicillin-Binding Proteins (PBPs)

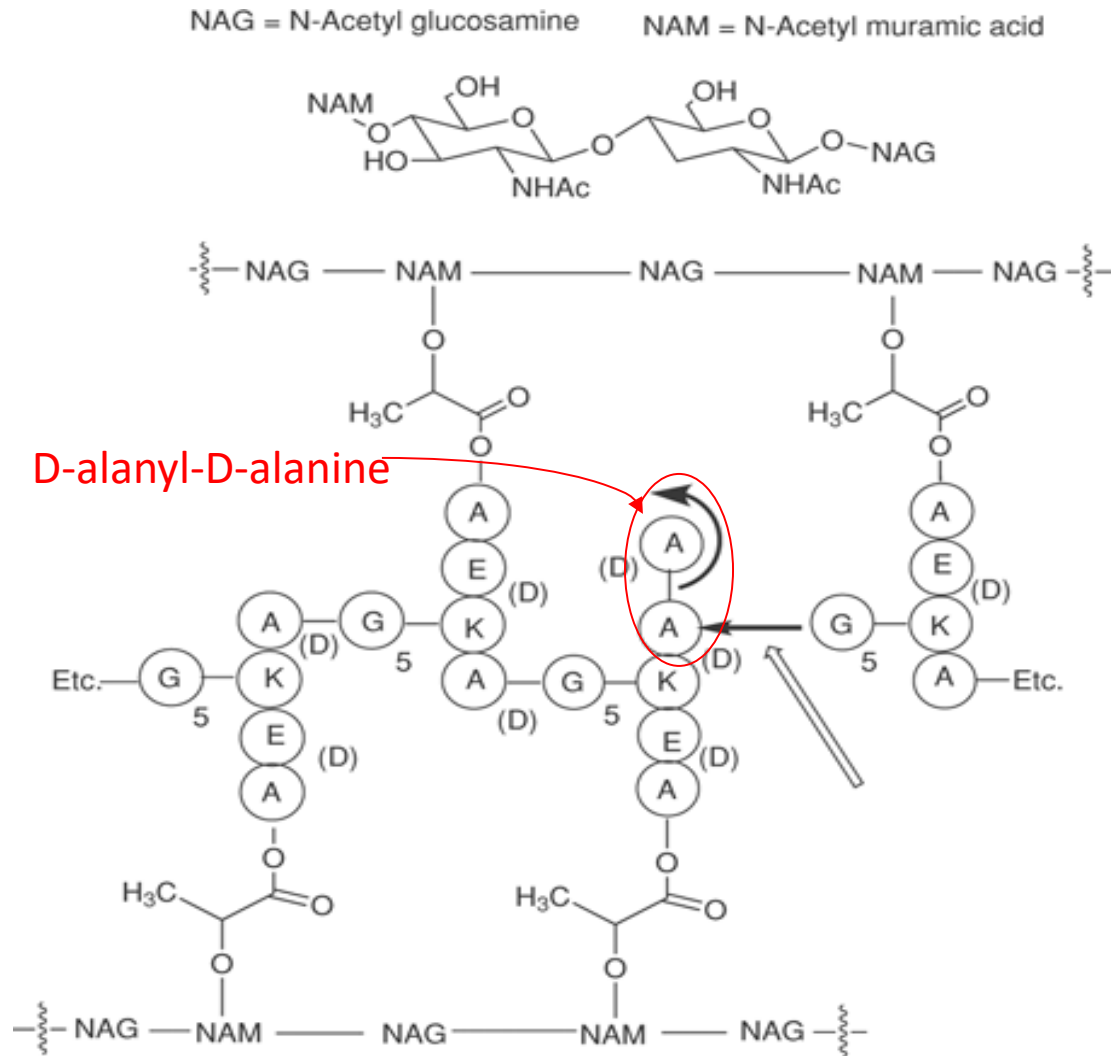
- PBPs are a set of proteins that have affinity for penicillins
- The name indicates the way by which the proteins were discovered
- All beta-lactam antibiotics bind to PBPs and prevent cell wall synthesis in bacteria

There are different membrane-bound PBPs that have different functions:

- PBPs 1a, 1b and 1c (most important): transpeptidases involved in peptidoglycan synthesis associated with cell elongation
- PBP 2: transpeptidase involved in maintaining the rod shape of bacilli
- PBP3: transpeptidase required for septum formation during cell division
- PBPs 4-6 (least important clinically): carboxypeptidases that limit the amount of cross-linking

Mechanism of Action of Penicillins

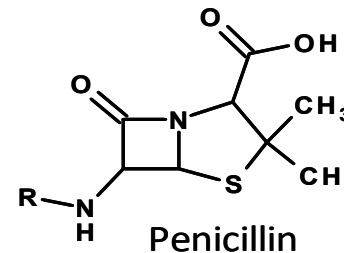
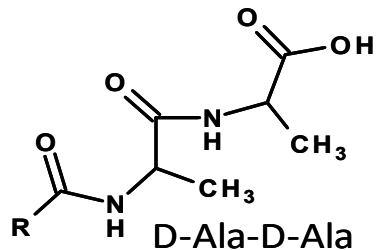
- Penicillins have a structure that closely resembles that of acylated D-alanyl-D-alanine
- The enzyme mistakenly recognizes the penicillin as though it were its normal substrate
- The highly strained β -lactam ring is much more reactive than a normal amide moiety
- After reacting with the penicillins, the enzyme's active site does not regenerate because of steric hindrance
- Inhibition of cross-linking results in structurally weak cell wall and the bacteria is subject to osmotic stress

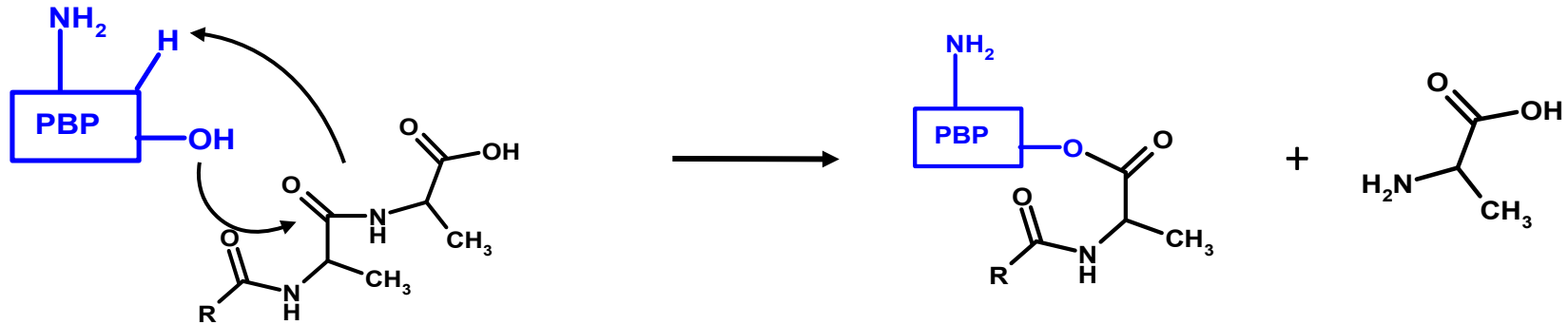


cell wall cross-linking: Pentaglycyl group replaces terminal D-alanine

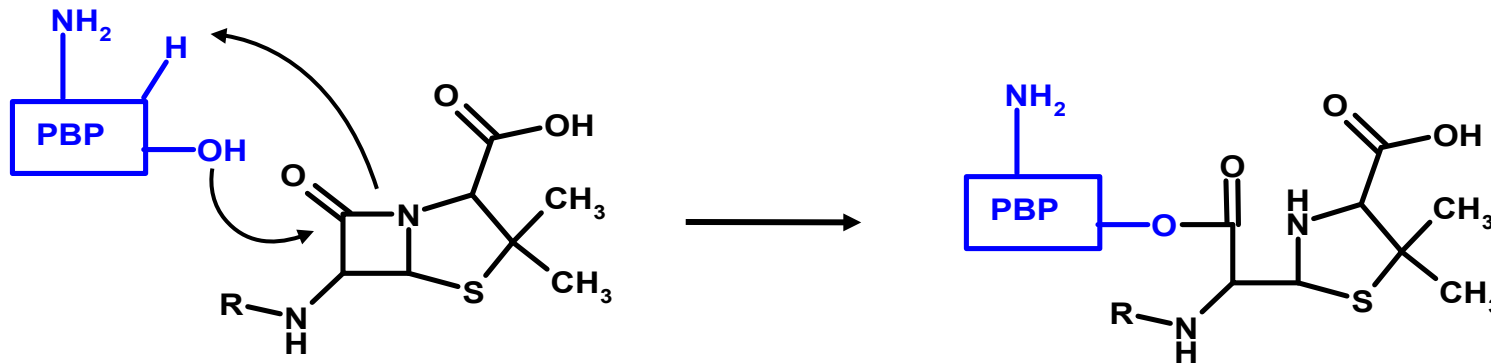
Mechanism of Action of Penicillins

- Penicillins inhibit PBPs that crosslink NAM peptide chains in different strands with pentaglycine units; inhibit peptidoglycan crosslinking
- Penicillin conformation is similar to the transition-state conformation adopted by the D-Alanine-D-Alanine moiety during the crosslinking reaction
- PBP mistakes penicillin for D-Alanine-D-Alanine and the β -lactam carbonyl groups is subjected to nucleophilic attack by the PBP active site serine residue
- Penicillin is cyclic, thus, the molecule is not split in two



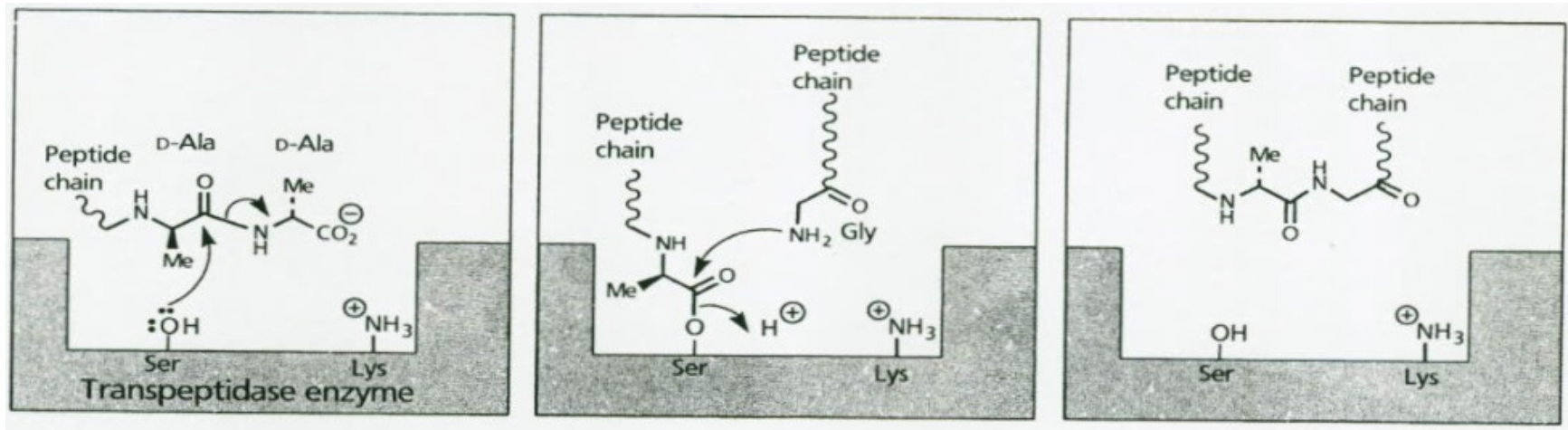


The mechanism of peptidoglycan crosslinking by PBP:
the terminal D-Alanine leaves the PBP active site. The acylenzyme intermediate can react with the pentaglycine chain to complete crosslinking

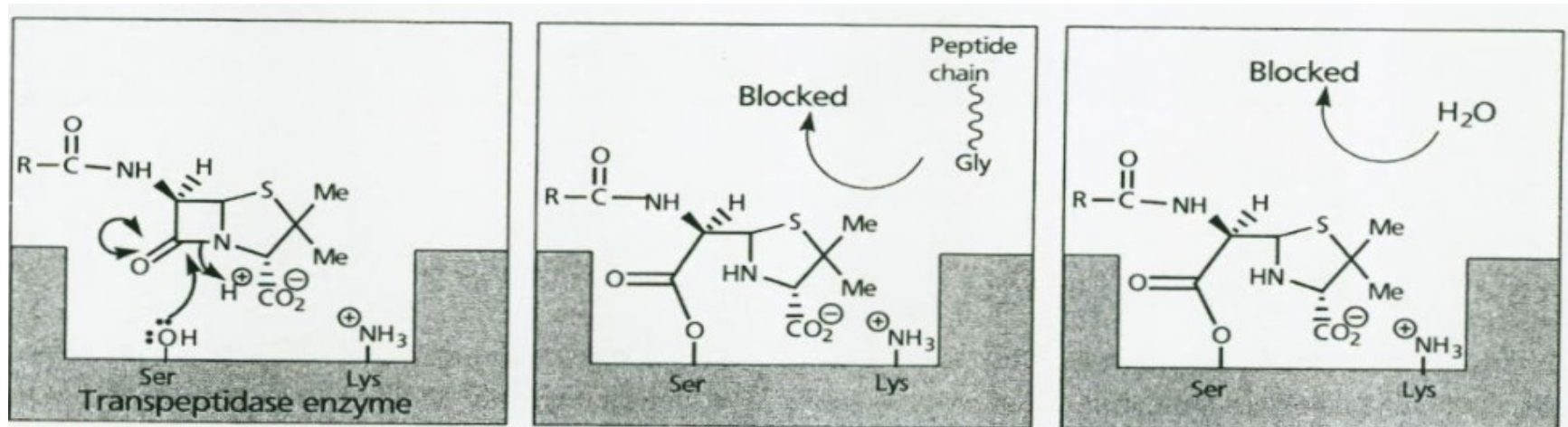


The mechanism of action of penicillin:
penicillin is cyclic; after nucleophilic attack penicillin is not split in two and nothing leaves the PBP active site. The penicillin structure remains at the active site and blocks access to the pentaglycine chain or water (nucleophiles)

Penicillin MOA



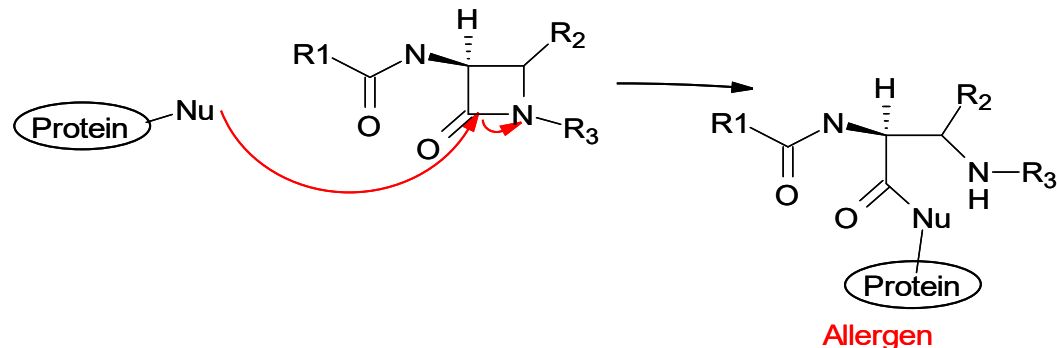
Mechanism of peptidoglycan crosslinking by PBP



The mechanism of action of penicillin

Allergenicity

- 6 - 8% of the U.S. population is allergic to β -lactam antibiotics
- mild drug rash or itching - most common
- immediate and profound reaction - seldom
- may include shock and cardiovascular collapse and even death
- medication history is important in anticipating penicillin allergy
- cross-sensitivity with other β -lactam is common
- the cause of the allergy is a haptenic reaction with host proteins and the β -lactam moiety
- **Small molecules that are not inherently antigenic but that react with proteins to produce antigens are called haptens**
- The **proteins serve as the nucleophiles** while the **β -lactam ring is an electrophile**



Allergenicity

- The group that is responsible for the allergic reaction is the β -lactam moiety
- Since the β -lactam moiety is an essential pharmacophore of the drug, separation of the allergenic property from its antibacterial activity has not been possible
- Penicillins may also be contaminated with small amounts of antigenic penicilloyl proteins derived from reaction with proteins encountered in their fermentative production
- Some penicillins may undergo self-condensation to give high-molecular weight polymers that can cause allergic reactions

Bacterial Resistance to Penicillins

- Production of β -Lactamase (penicillinase) enzymes
 - open up the β -lactam rings of β -lactams (penicillins) and inactivate them.
- Some Gram-negative bacteria produce increased levels of PBPs; thus, penicillin is incapable of inactivating all PBP enzyme molecules produced.
- Production of PBPs with lower affinity for penicillins
- In some Gram-negative bacteria, efflux of penicillins across the outer cell membrane

β -Lactamase Inhibitors

- Mechanism-based inhibitors
- Have β -lactam ring
- Have no intrinsic antibacterial activity

